

If a language is defined by any one of the (Th, FA, Regex) then it can be defined by other way as well.

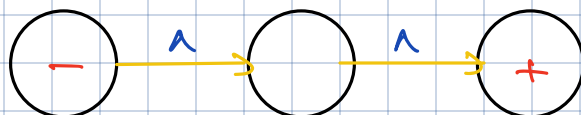
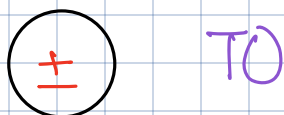
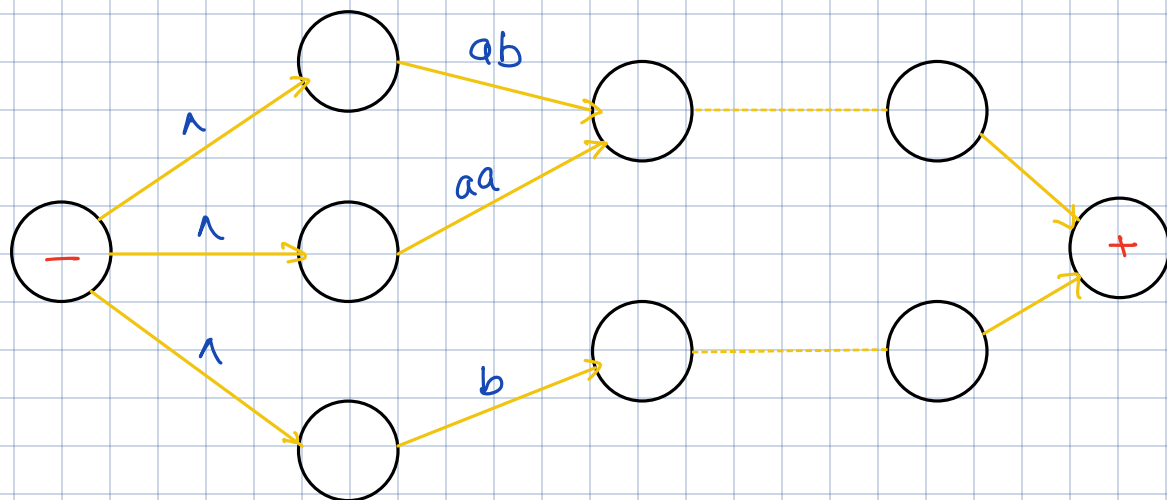
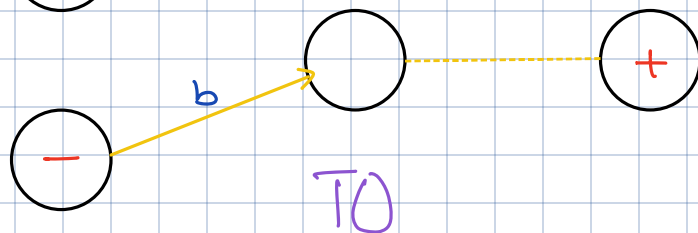
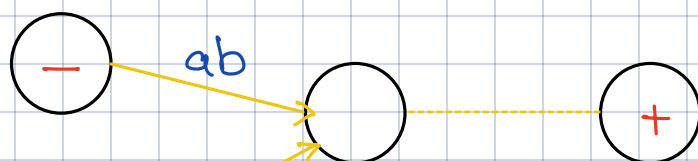
i.e All three of these ways of defining language are equivalent.

1. Every language defined by FA is defined by Th.

↳ Every FA is a Th

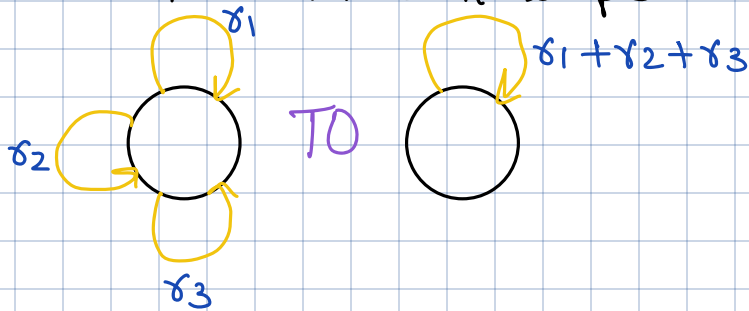
2. Every language defined by Th can be defined by R.E.

① Simplify Th by have only one start & final state.

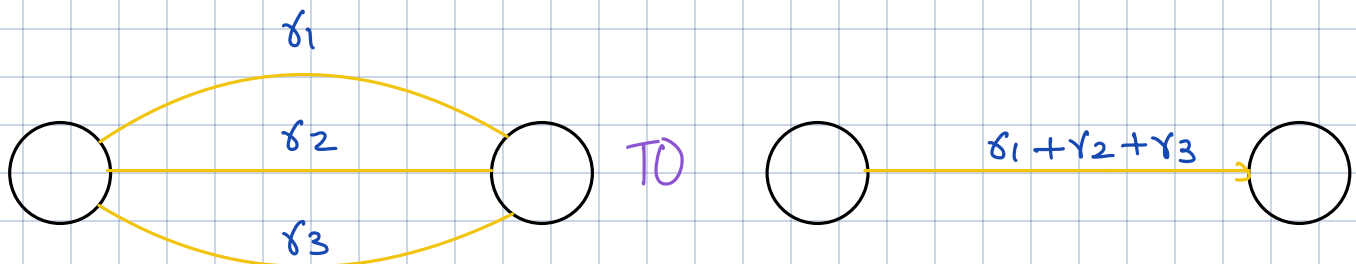
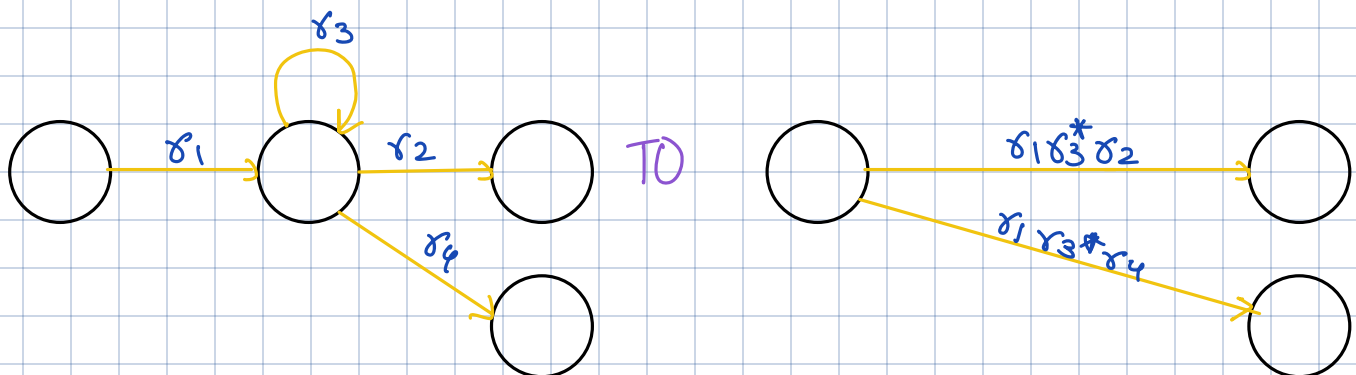
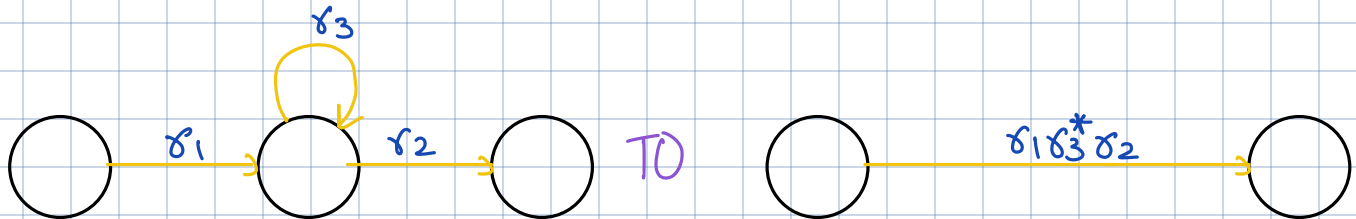


② Build R.E by eliminating states

- Deal with self loops

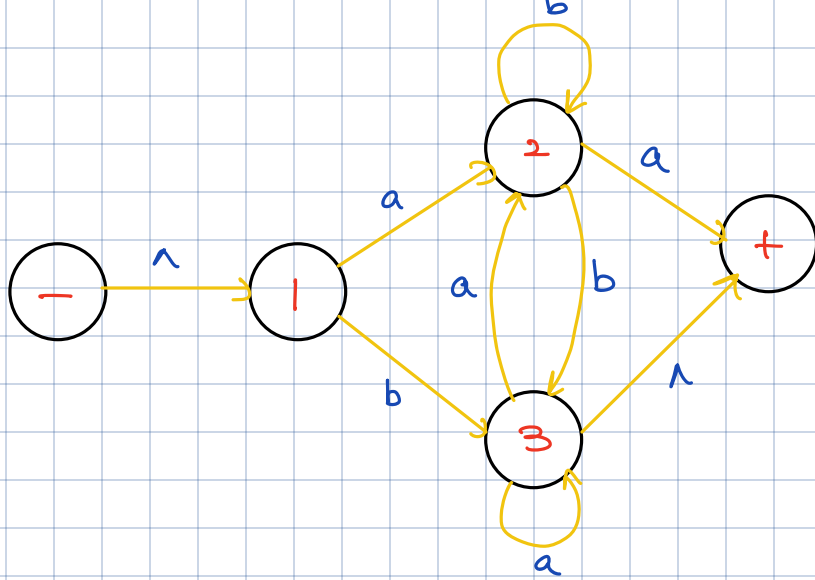


- Eliminate states



↳ Do this repeatedly till you are left with only - $\times$ +.

Example



Eliminate 1, 2, 3

For loop,

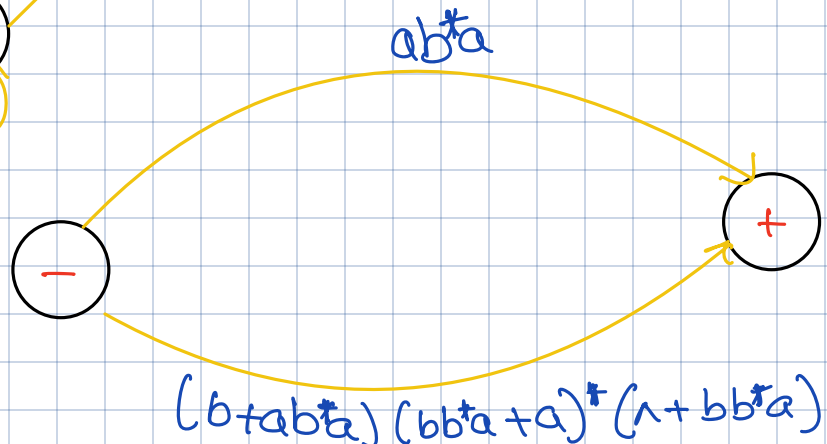
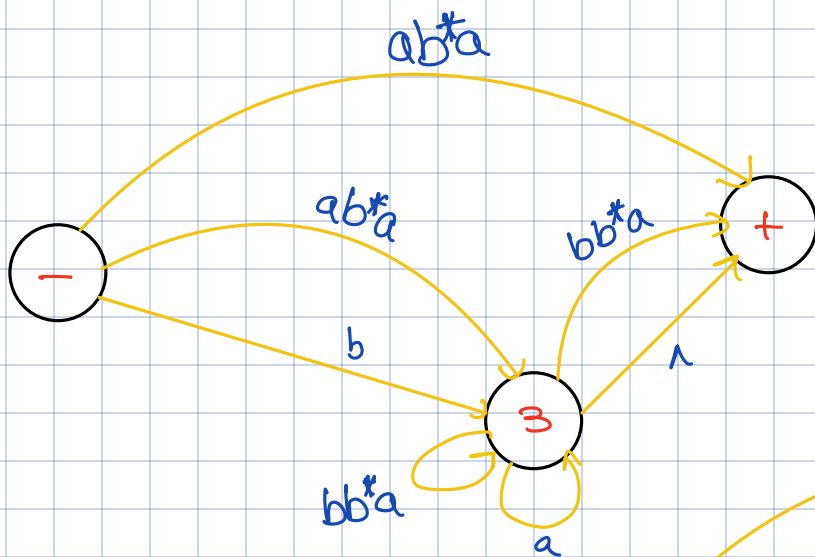
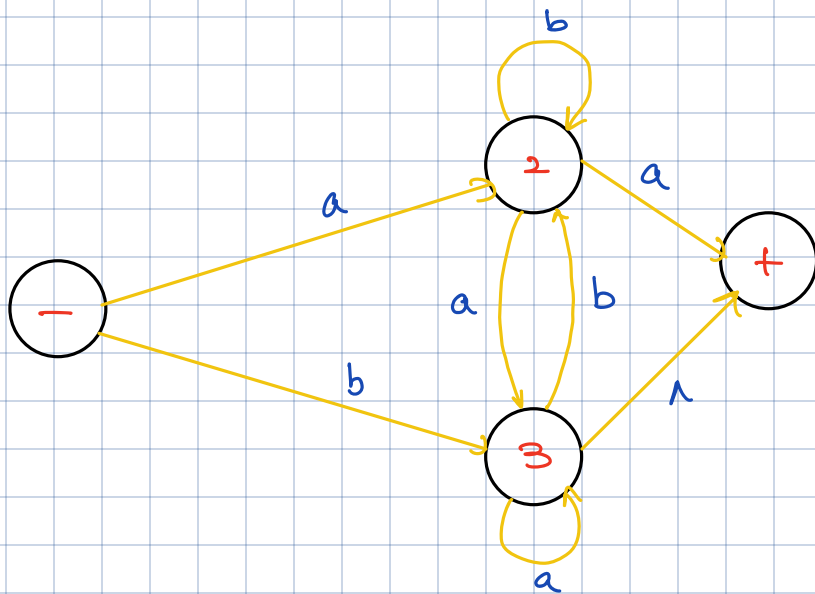
Create regex self on the other state.

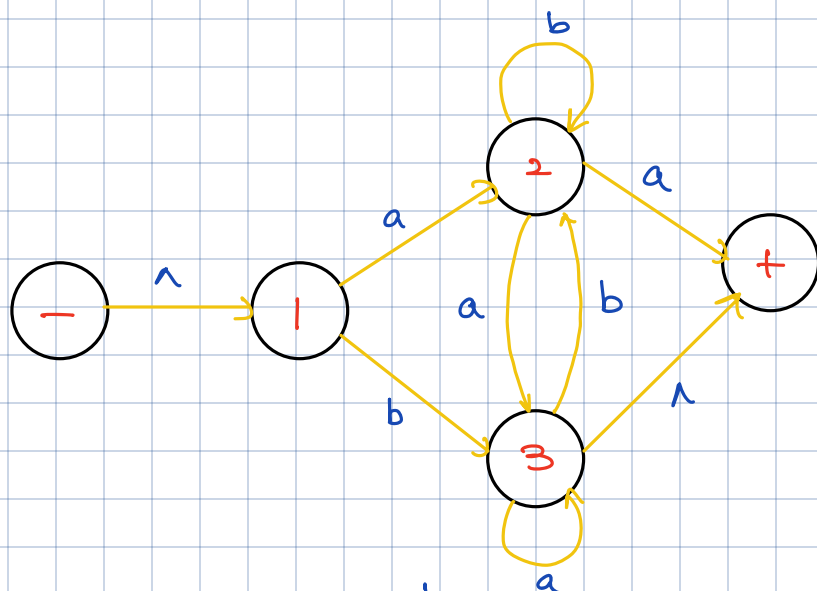
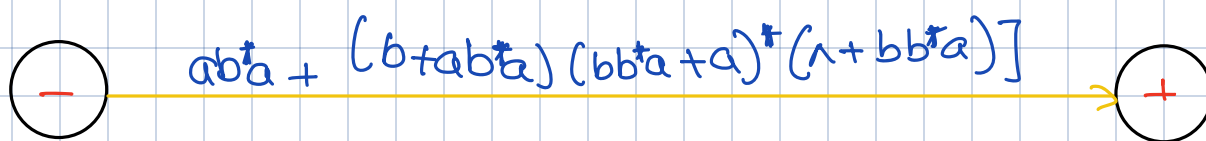
regex should be from going to eliminated state

for example,

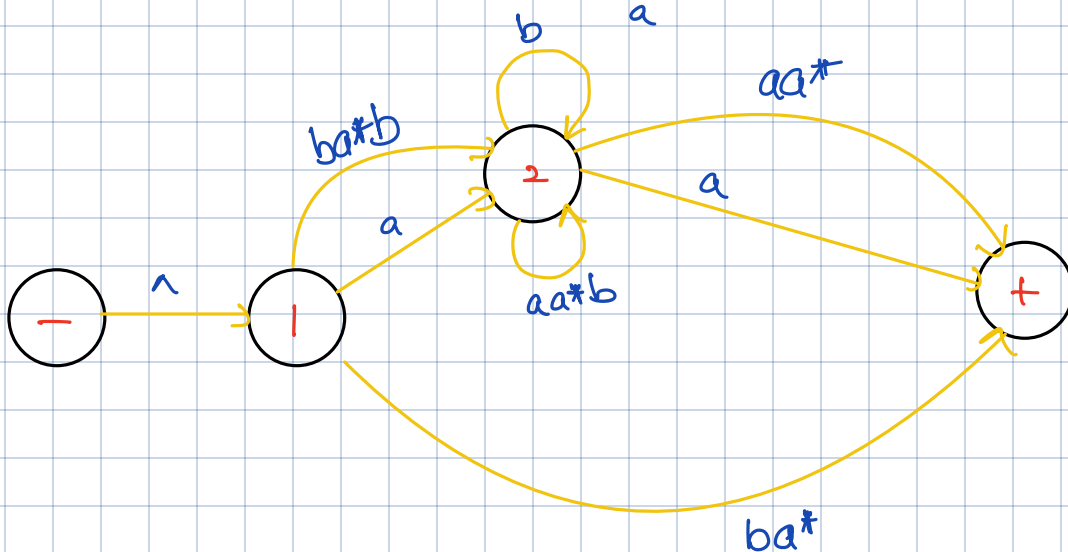
abba and NOT baab.

Covers all paths via the eliminated states





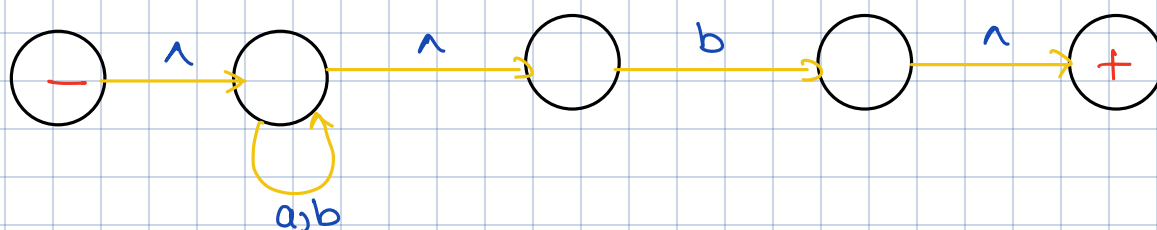
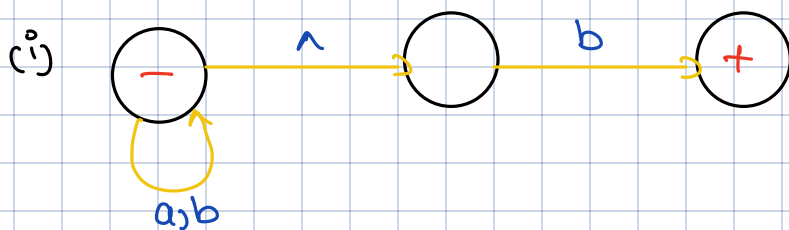
eliminate 3, 2, 1

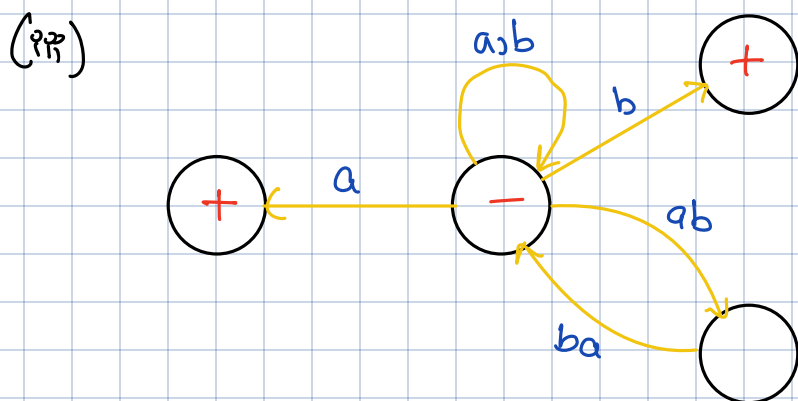
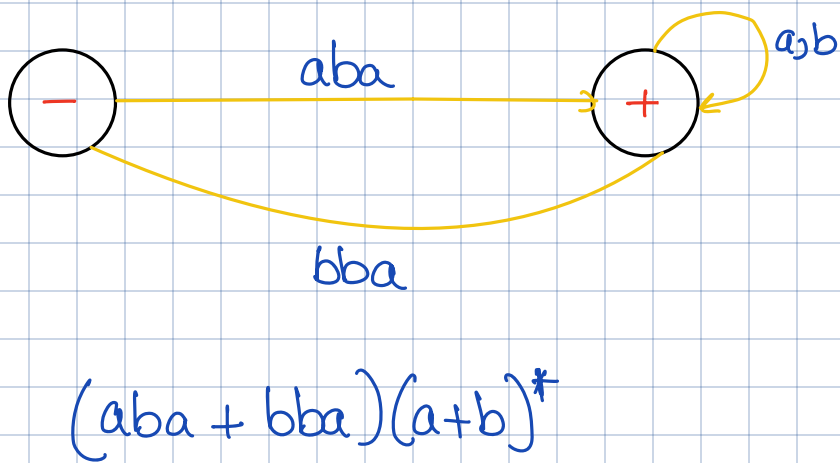
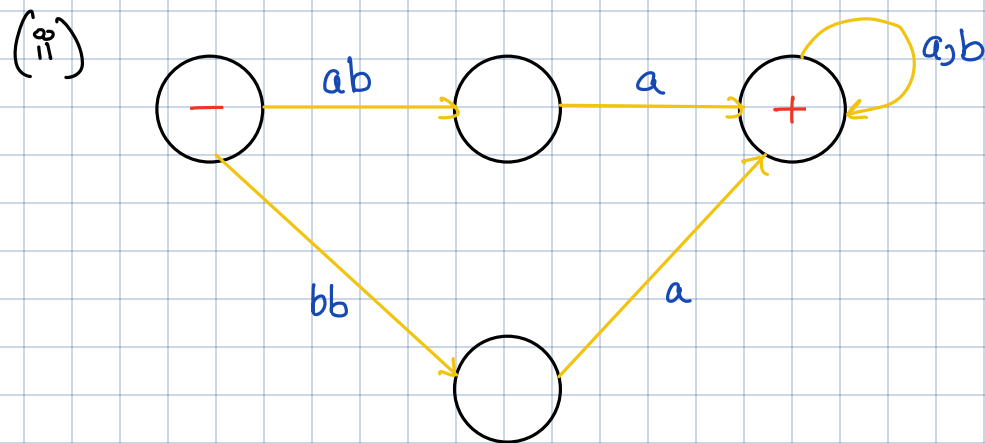
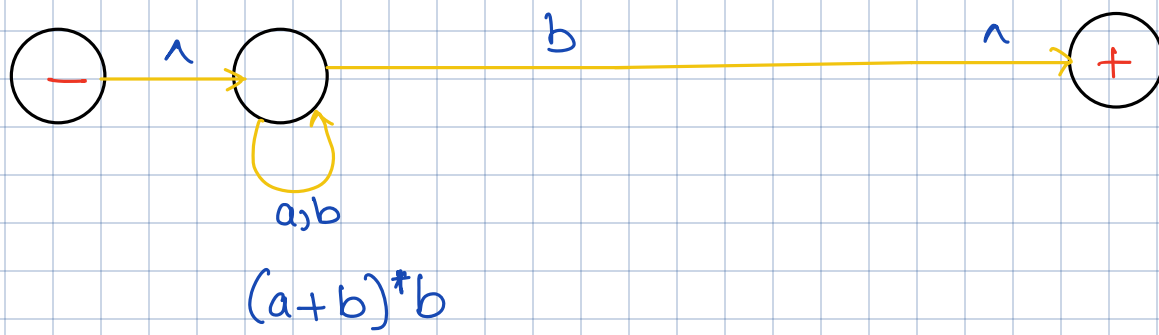


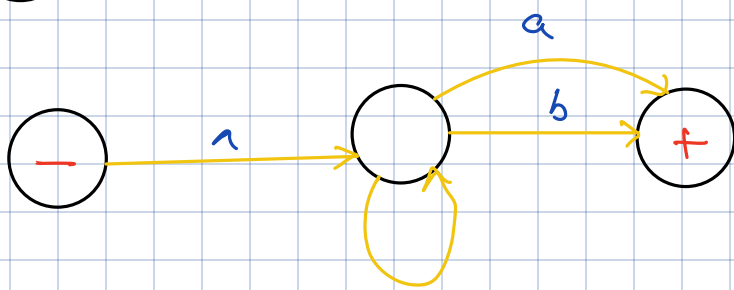
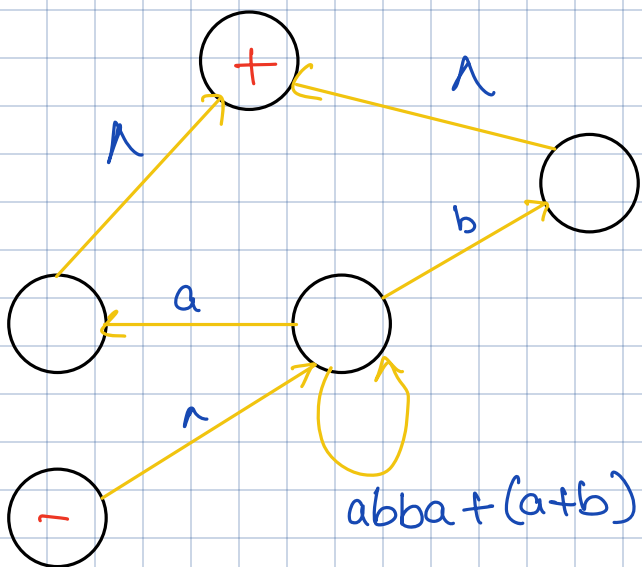
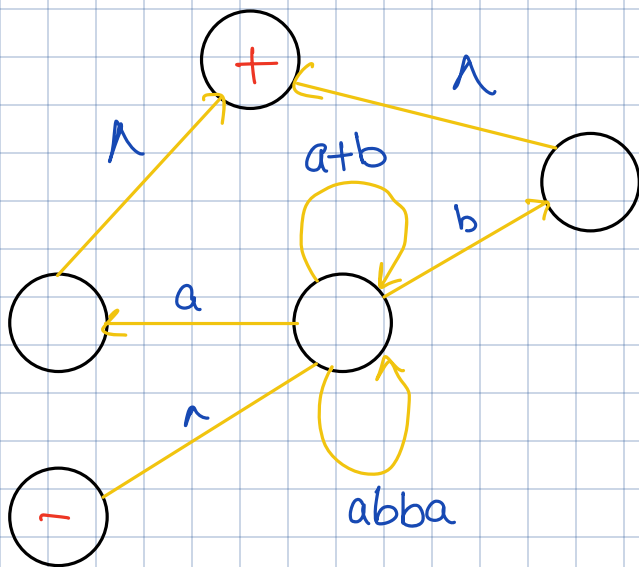
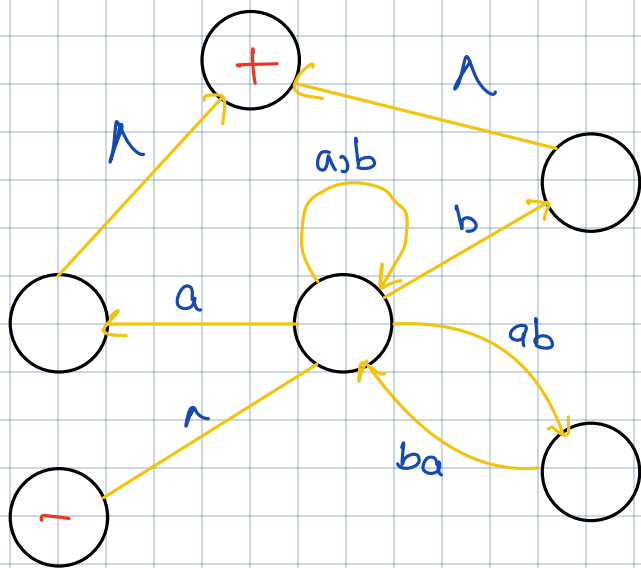
$$[(a + ba^*b)(b + aa^*b)^*(a + aa^*)] + ba^*$$

Problems

Q1.



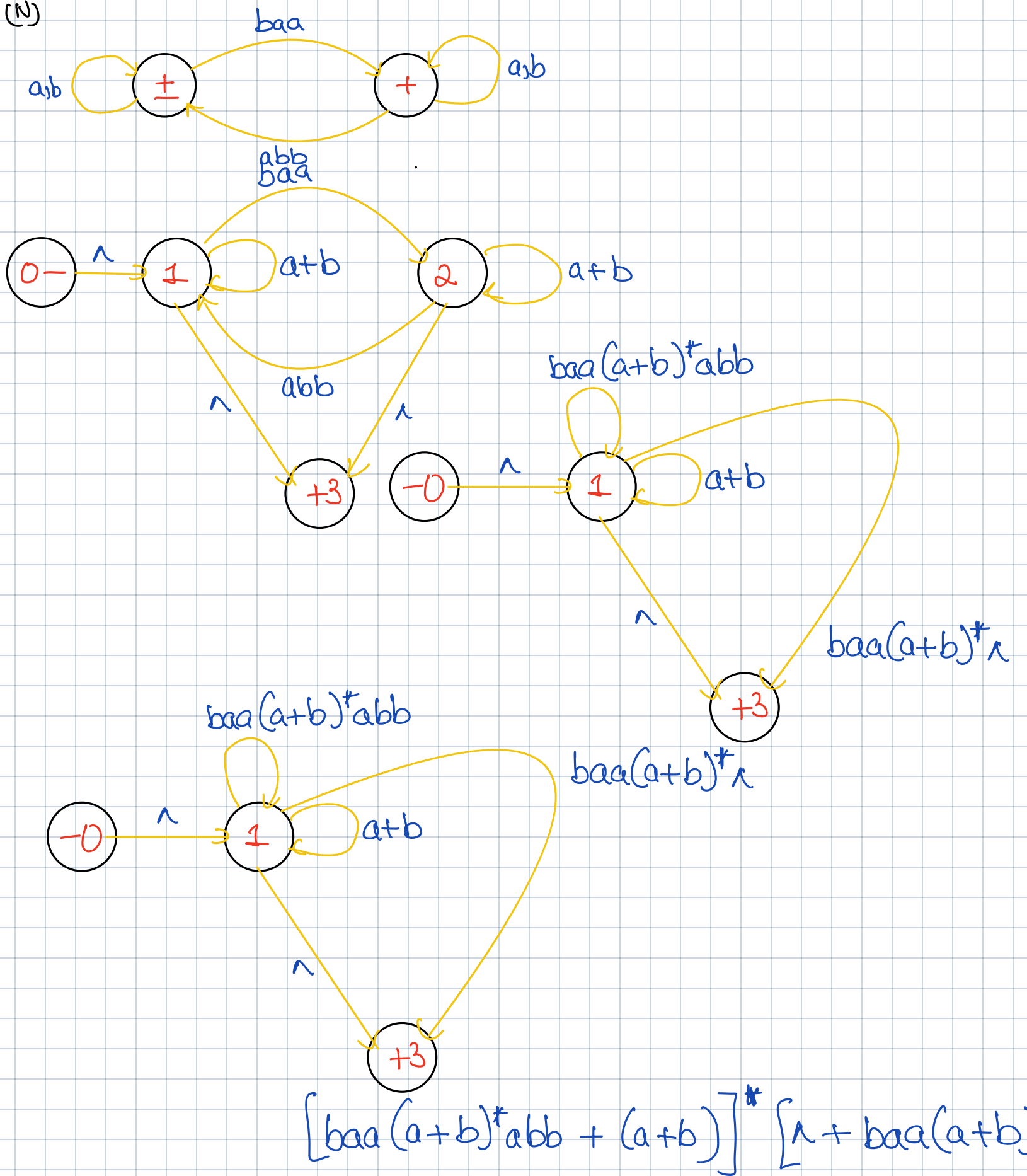




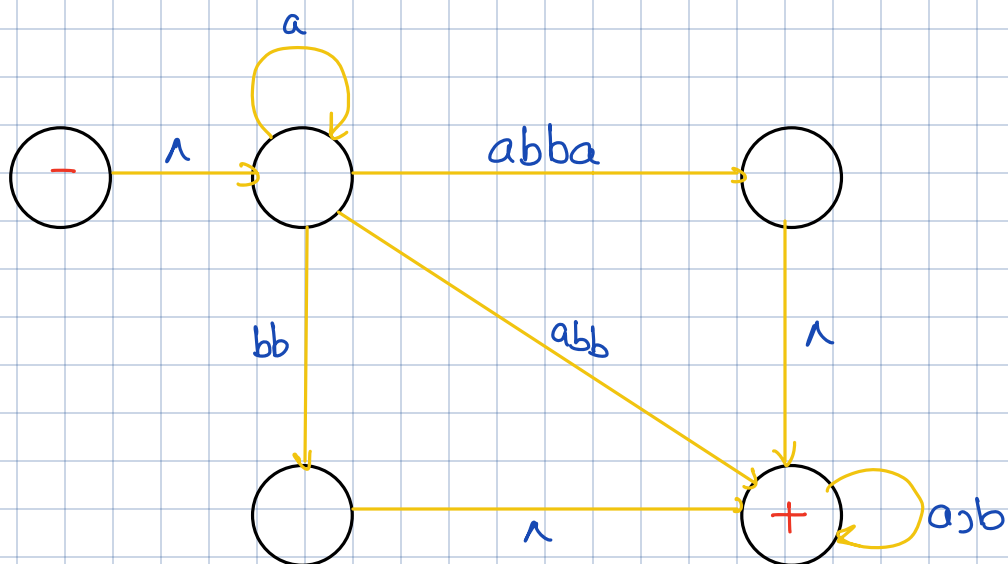
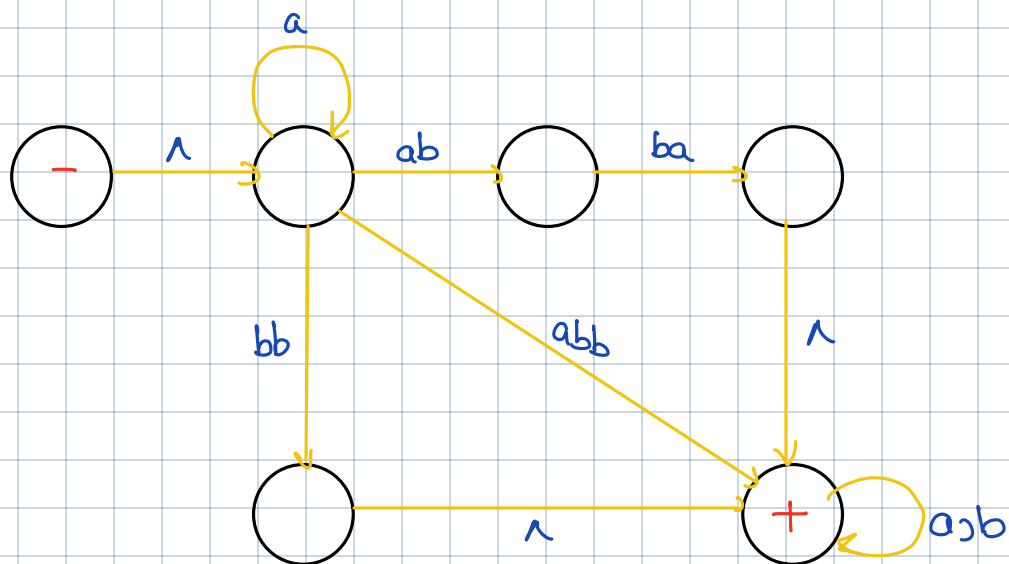
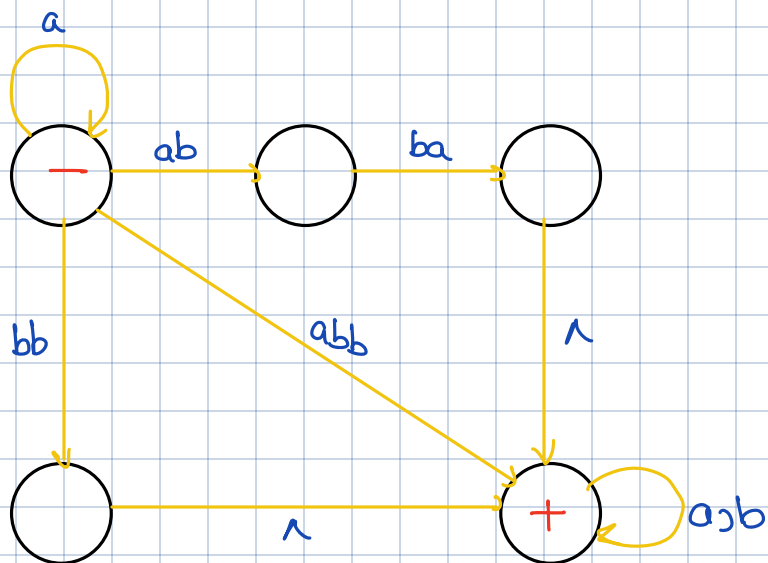
$$abba + (a+b)$$

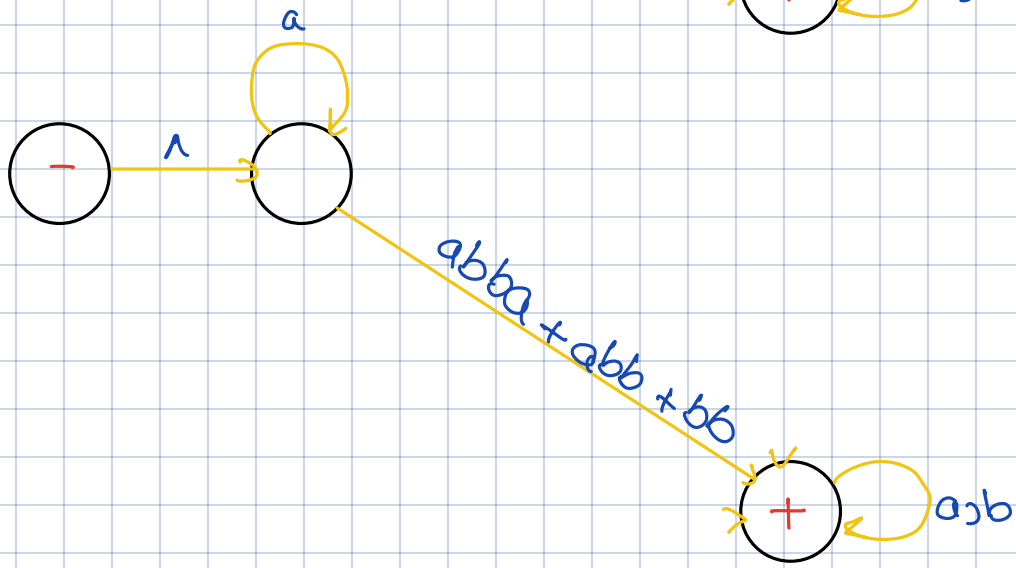
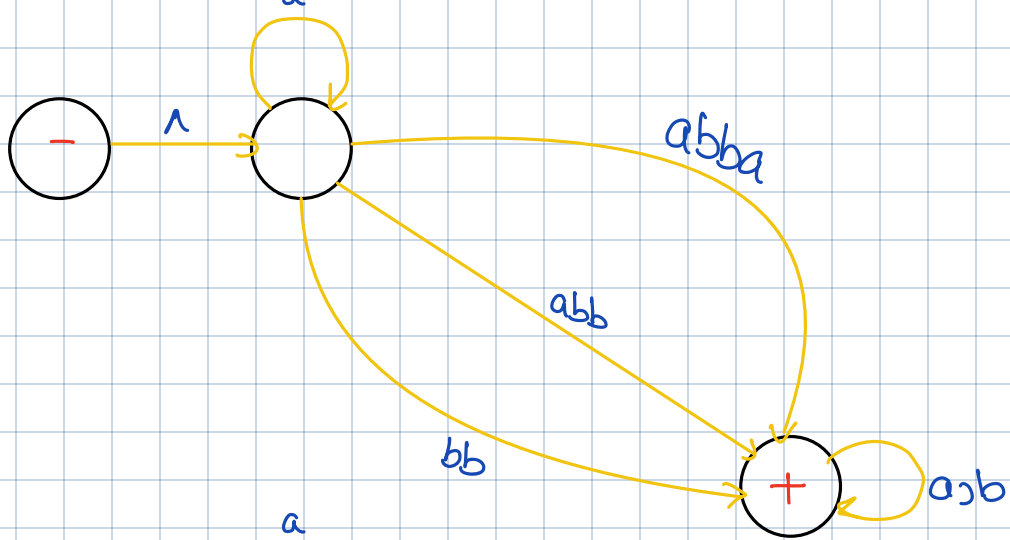
$$[abba + (a+b)]^*(a+b)$$

(N)



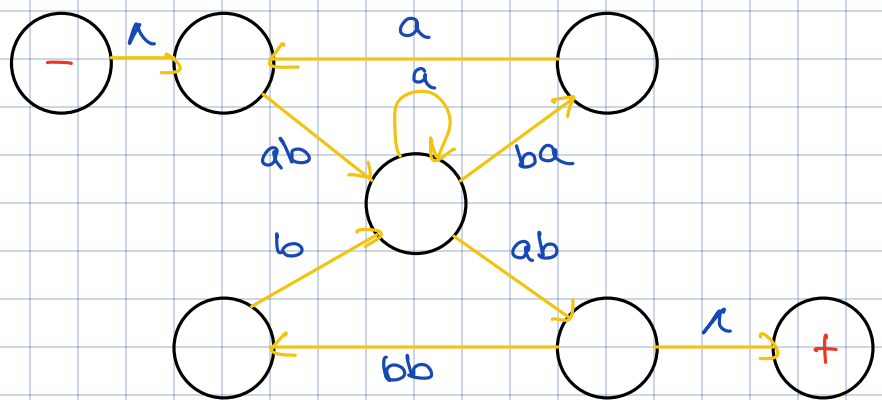
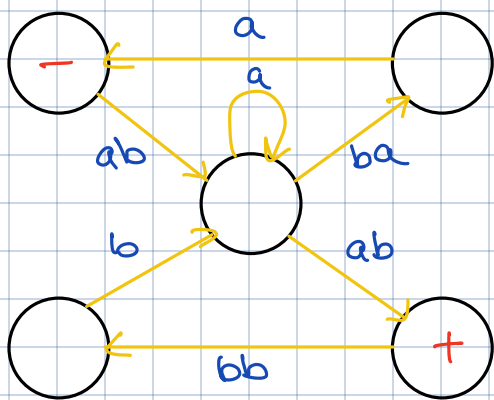
(V)

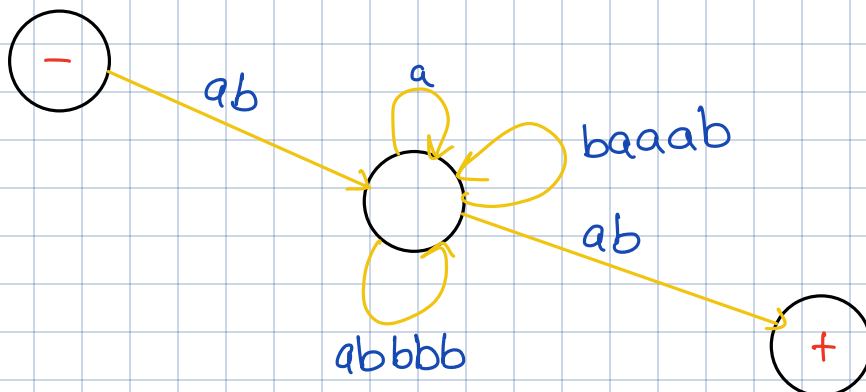
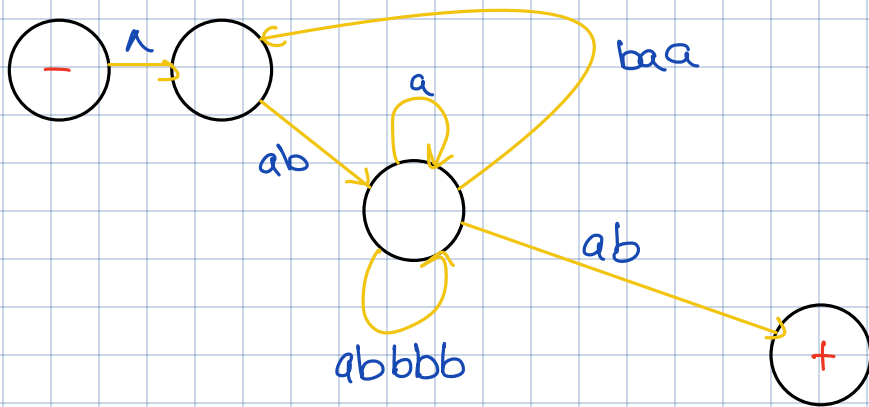
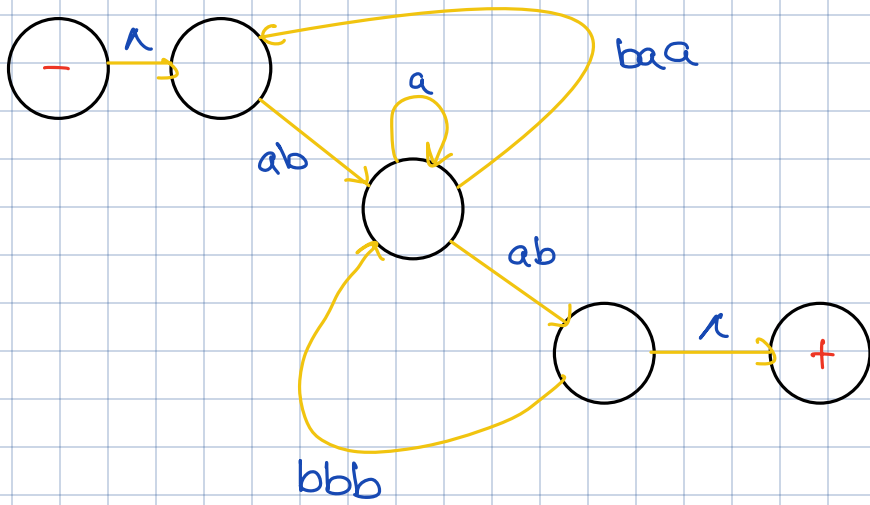
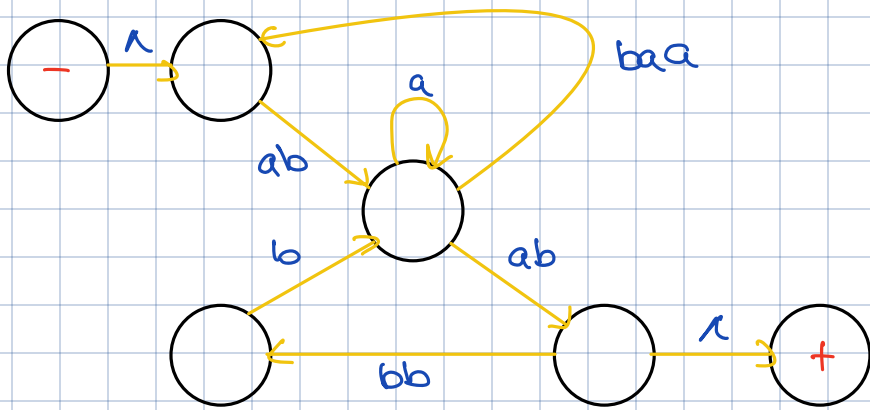




$$a^*(abba+abb+ab)(a+b)^*$$

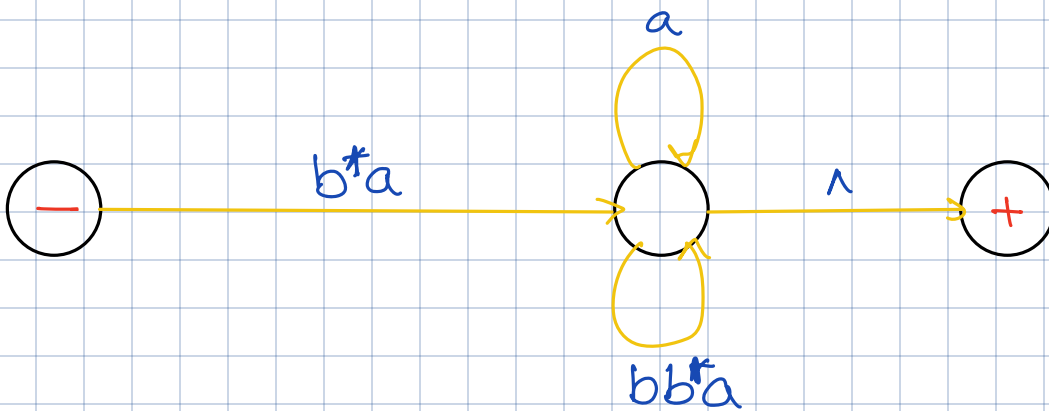
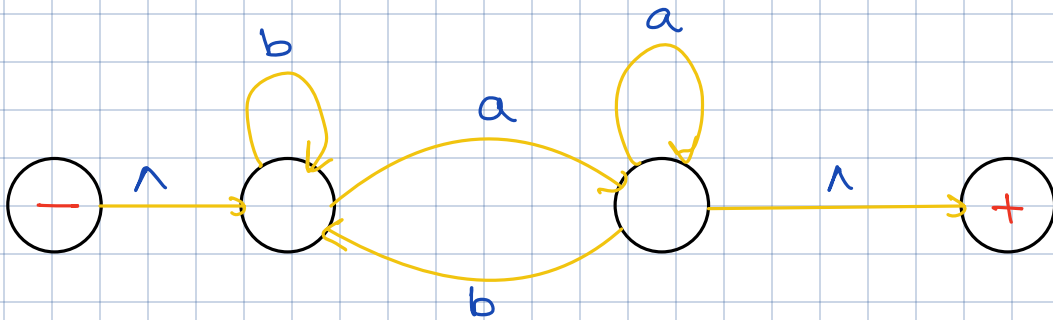
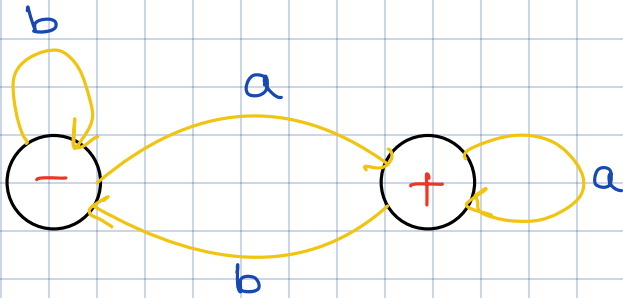
(vii)



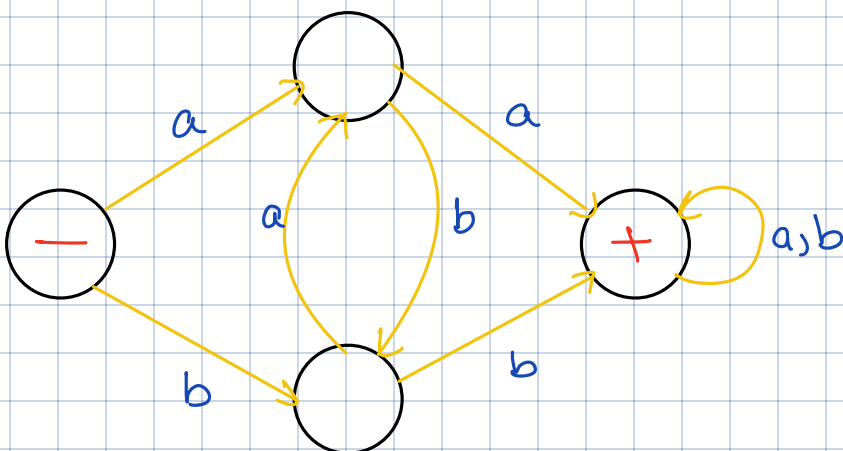


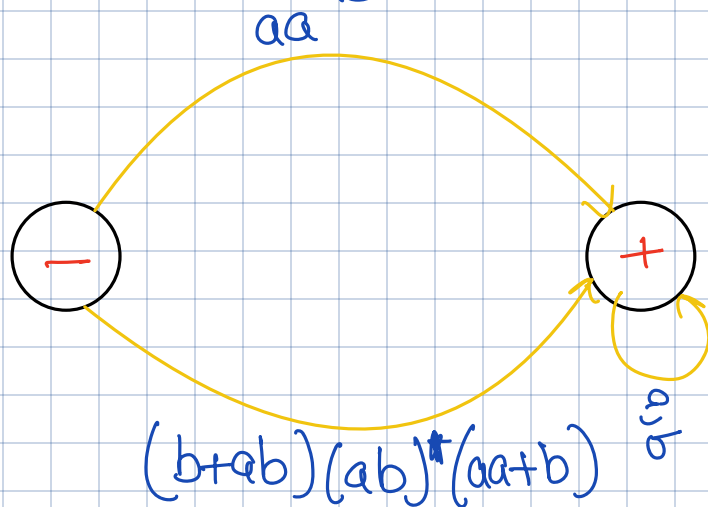
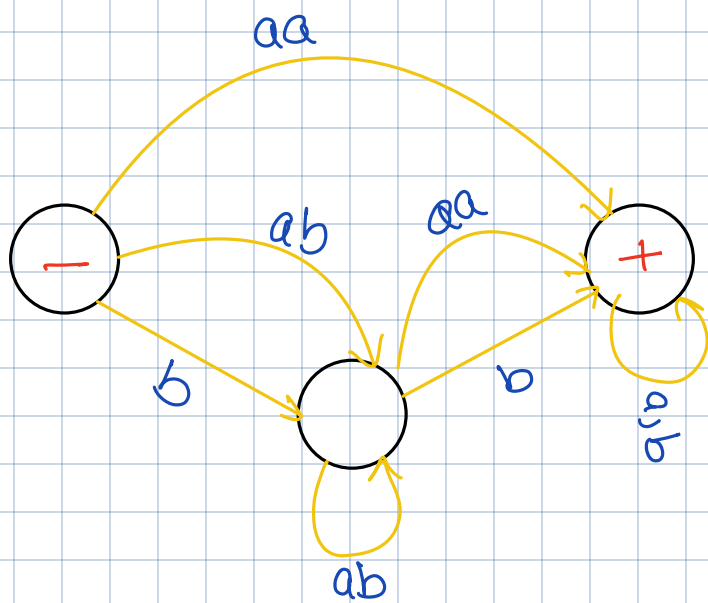
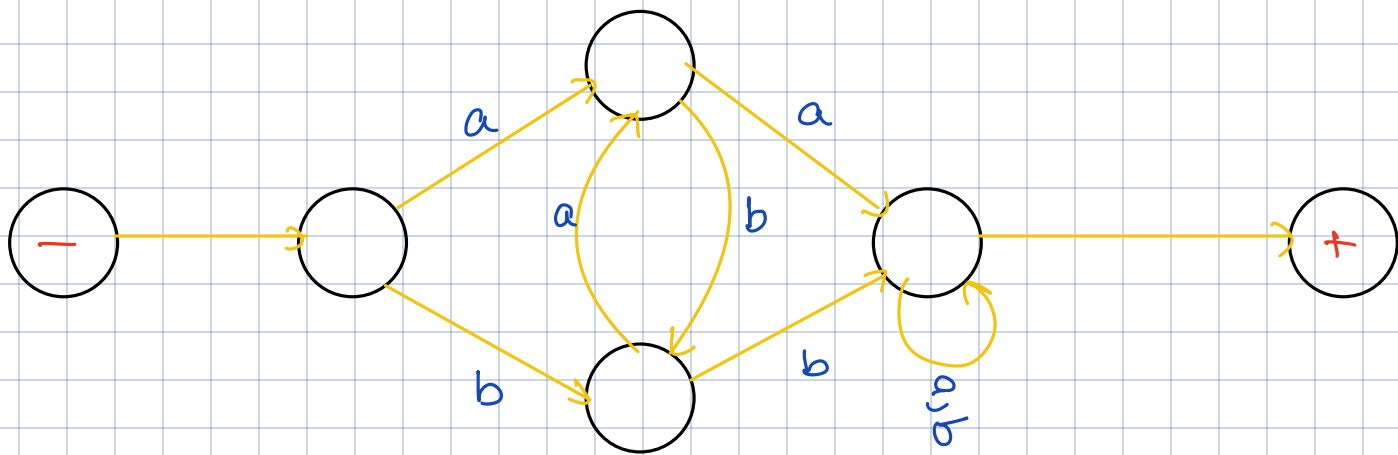
$$ab[a^* + (baab)^* + (abbbb)^*]ab$$

Q2.

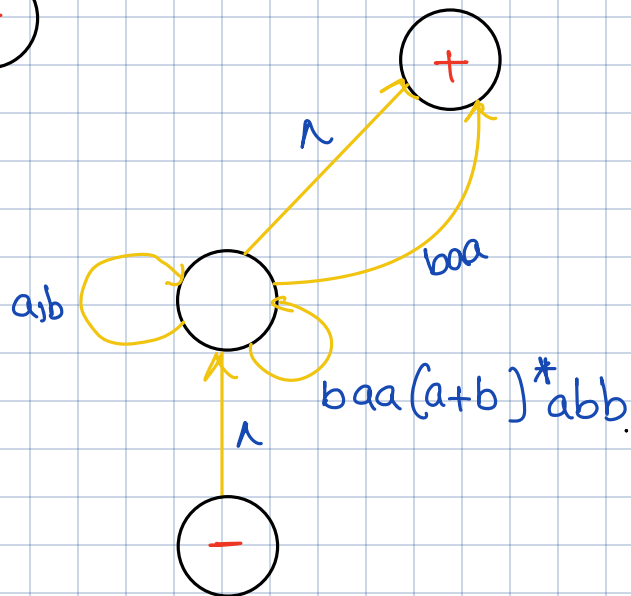
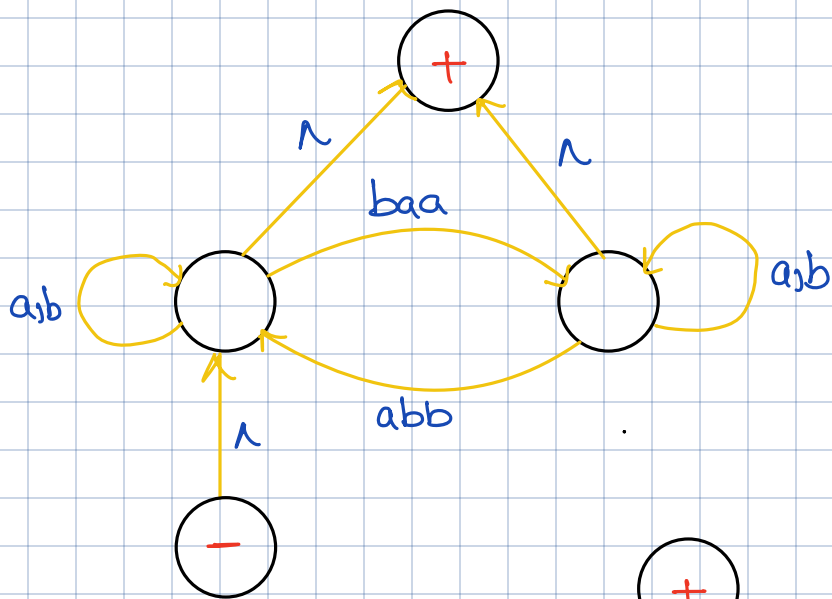


$$b^*a[a^* + bb^*a]$$





$$\left[aa + (b+ab)(ab)^*(aa+b) \right] (a+b)^*$$



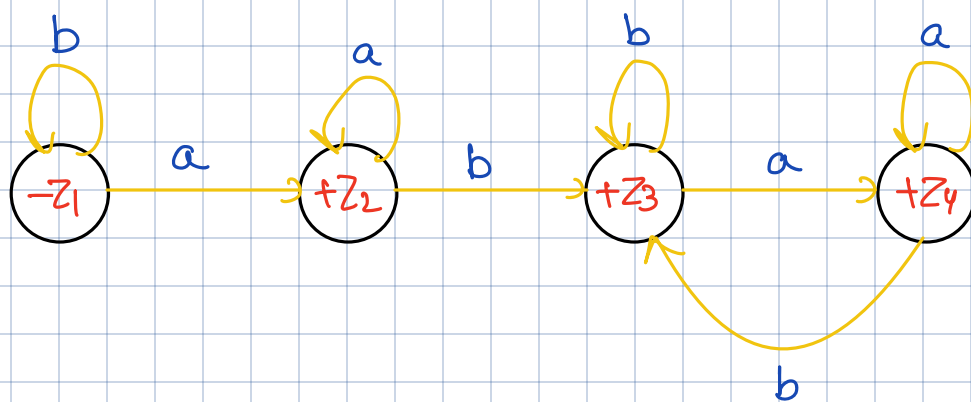
$$[(a+b) + (baa(a+b)^*abb)]^* [a + bba]$$

Q3.

(i) $FA_1 + FA_2$

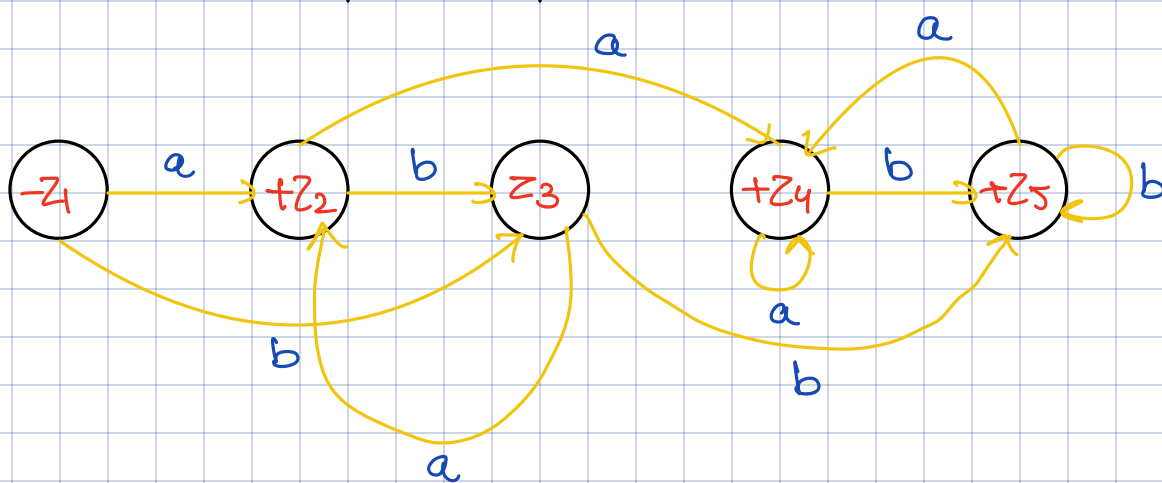
	a	b
$Z_1 = -w_1 0x - x_1$	$+Z_2$	$-Z_1$
$+Z_2 = +w_2 0x + x_2$	$+Z_2$	$+Z_3$
$+Z_3 = -w_1 0x + x_3$	$+Z_4$	$+Z_3$
$+Z_4 = +w_1 0x + x_4$	$+Z_4$	$+Z_3$

$$+Z_4 = +w_2 \cdot y_3 + Z_3$$



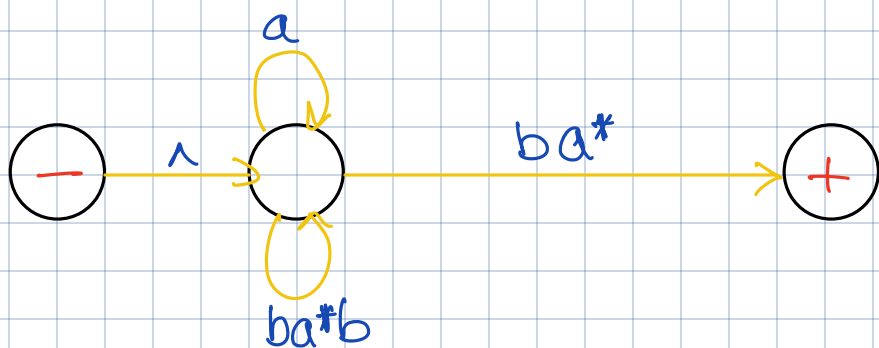
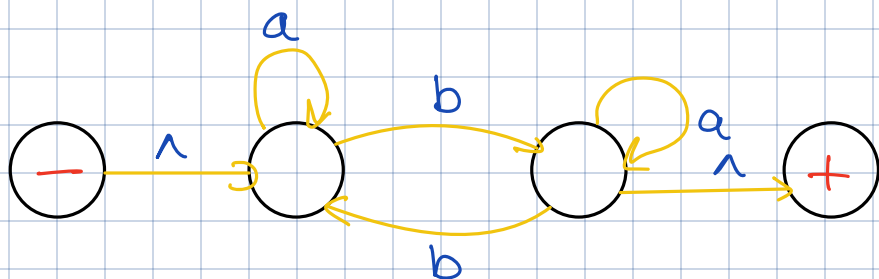
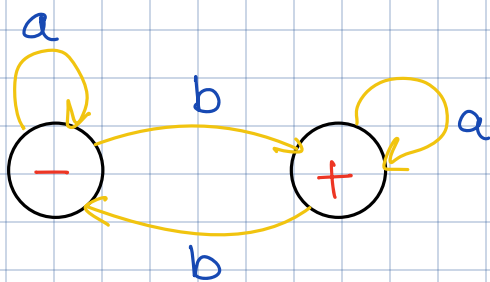
ii) $FA_1 + FA_3$

	a	b
$Z_1 = -w_1 \cdot x - y_1$	$+Z_2$	Z_3
$+Z_2 = +w_2 \cdot y_2$	$+Z_4$	Z_3
$Z_3 = -w_1 \cdot x + y_3$	$+Z_2$	Z_5
$+Z_4 = +w_2 \cdot x + y_4$	$+Z_4$	Z_5
$+Z_5 = -w_1 \cdot x + y_4$	$+Z_4$	$+Z_5$

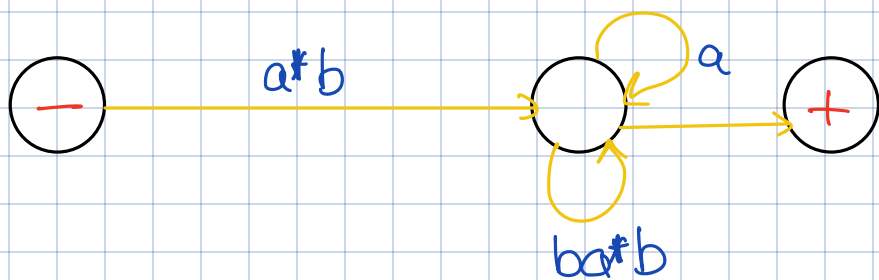


Q1.21

(a)

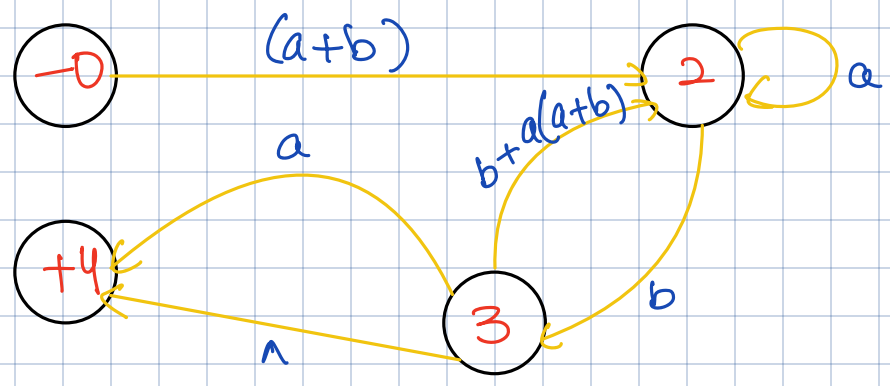
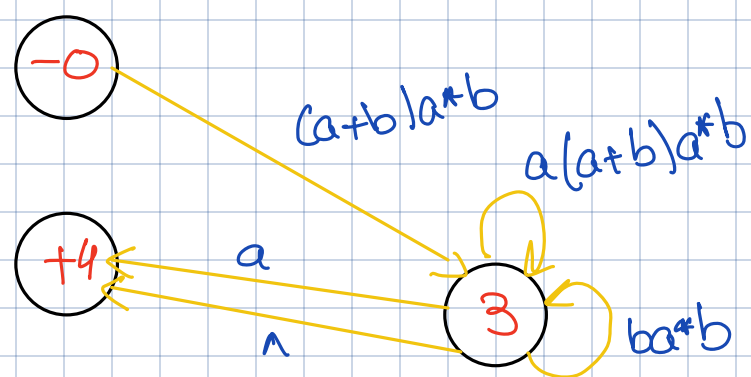
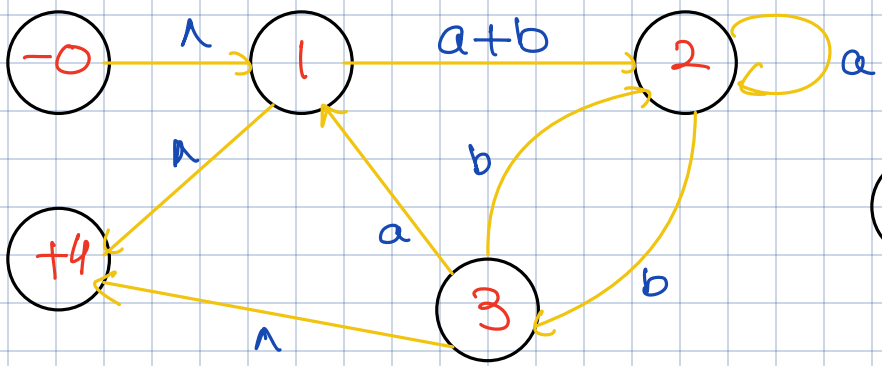
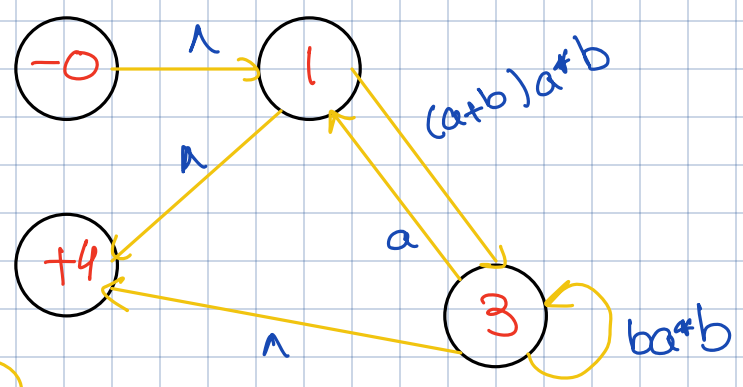
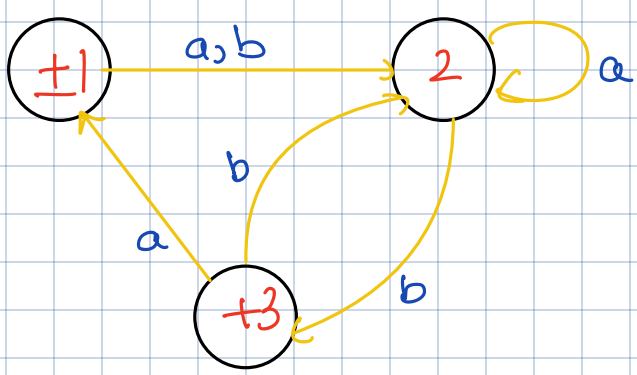


$$(a + ba^*b)^* ba^*$$

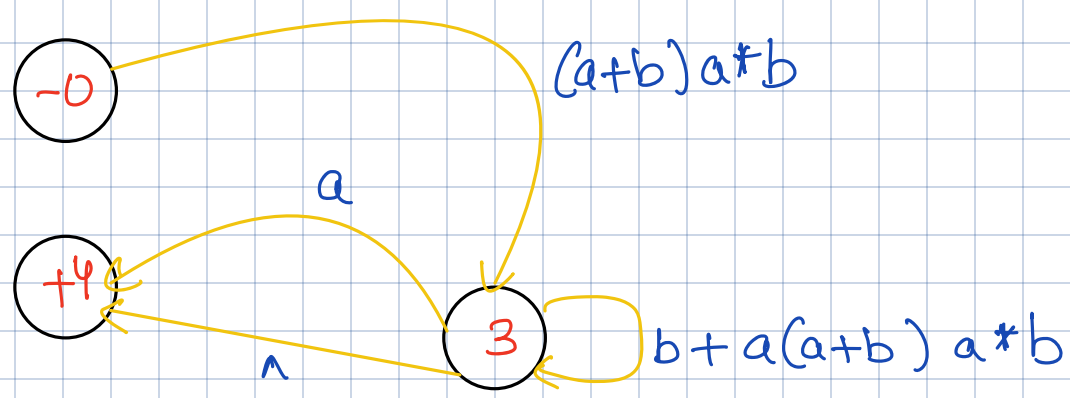


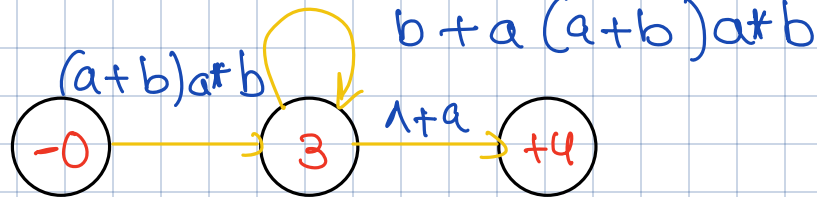
$$a^*b(ba^*b + a)^*$$

(b)



$$(a+b)a*b \left[a(a+b)a*b + ba*b \right]^* (\Lambda + a)$$



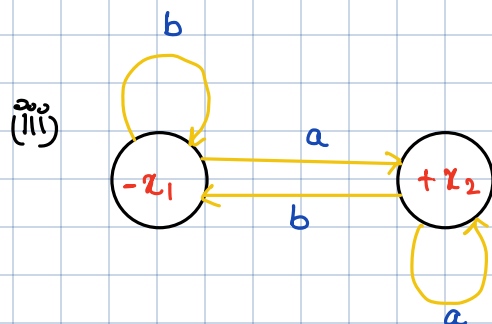
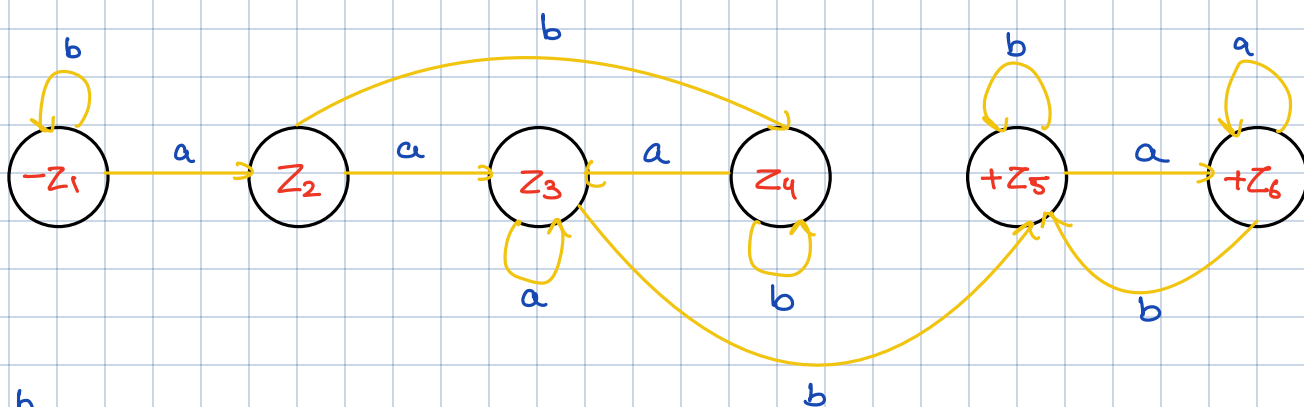


$$(a+b)a^*b [b+a(a+b)a^*b]^* (1+a)$$

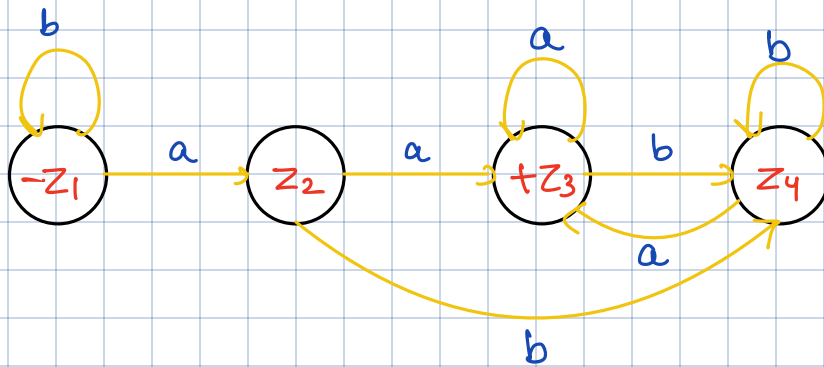
Q5.

(i) FA₁ FA₂

	a	b
$-Z_1 = w_1$	Z_2	Z_1
$Z_2 = w_2 \circ x_1$	Z_3	Z_4
$Z_3 = w_2 \circ x_1 \circ x_2$	Z_3	$+Z_5$
$Z_4 = w_1 \circ x_1$	Z_3	Z_4
$+Z_5 = w_1 \circ x_1 \circ x_3$	$+Z_6$	Z_5
$+Z_6 = w_2 \circ x_1 \circ x_2 \circ x_3$	$+Z_6$	$+Z_5$

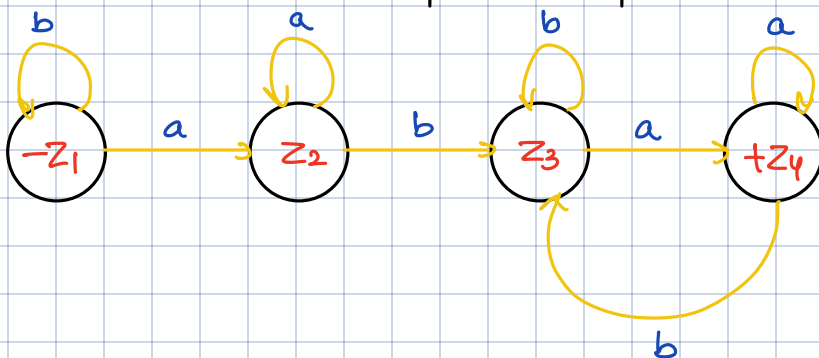


	a	b
$-Z_1 = \omega_1$	Z_2	Z_1
$Z_2 = \omega_2 \oslash \omega_1$	$+Z_3$	Z_4
$+Z_3 = \omega_2 \oslash \omega_1 \oslash \omega_2$	$+Z_3$	Z_4
$Z_4 = \omega_1 \oslash \omega_1$	$+Z_3$	Z_4



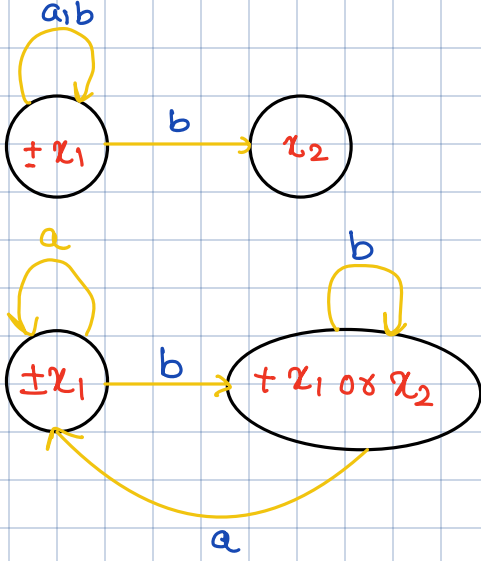
(iv) $FA_2 FA_1$

	a	b
$-Z_1 = \omega_1$	Z_2	Z_1
$Z_2 = \omega_2$	Z_2	Z_3
$Z_3 = \omega_3 \oslash \omega_1$	Z_4	Z_3
$+Z_4 = \omega_3 \oslash \omega_1 \oslash \omega_2$	Z_4	Z_3

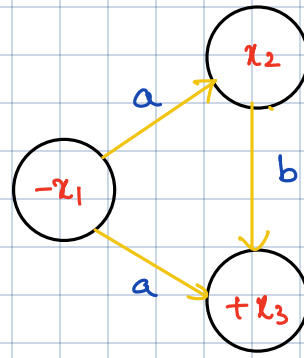


Q 14.

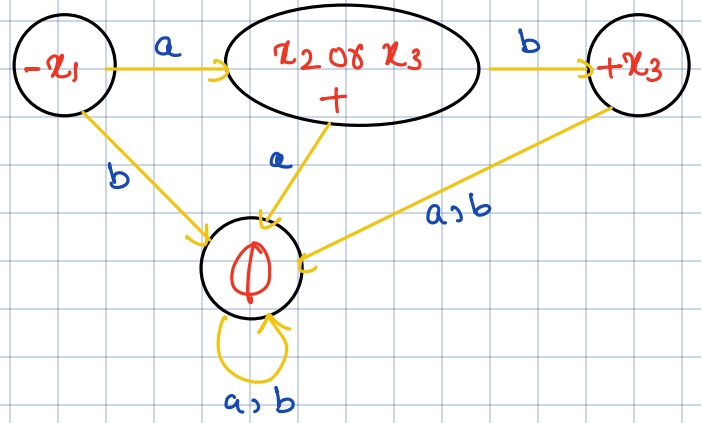
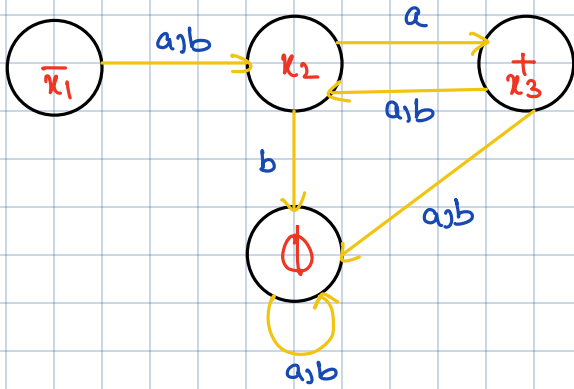
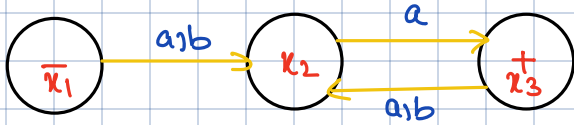
(i)



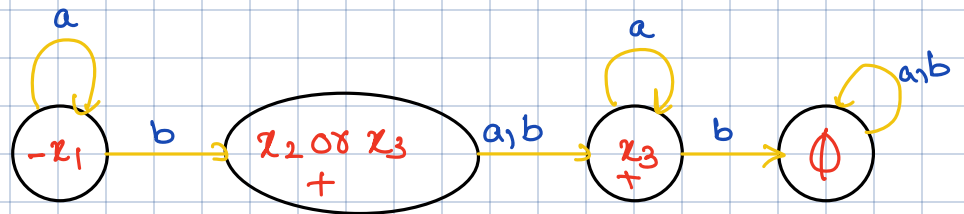
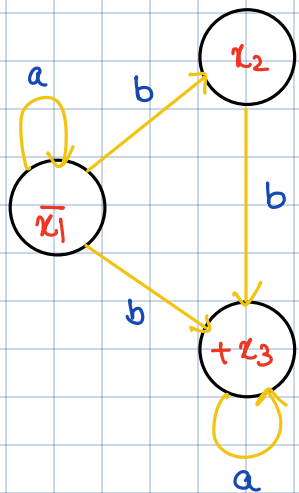
(ii)



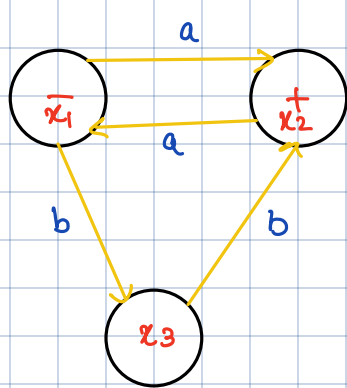
(iii)



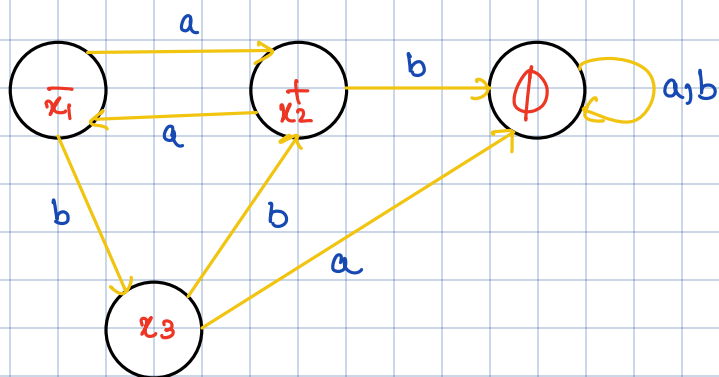
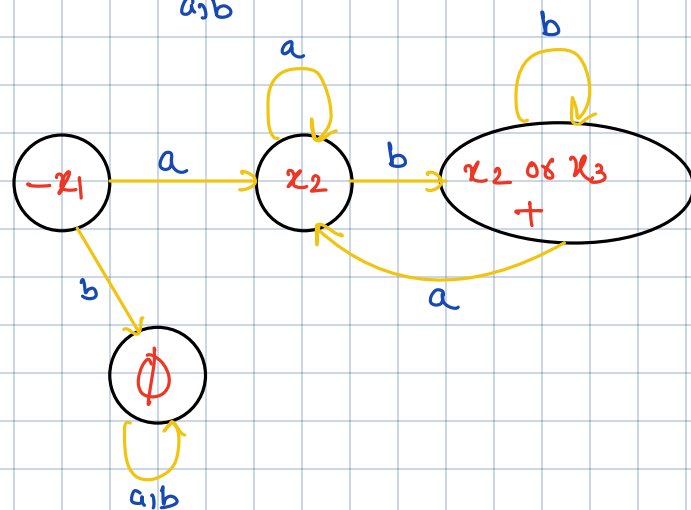
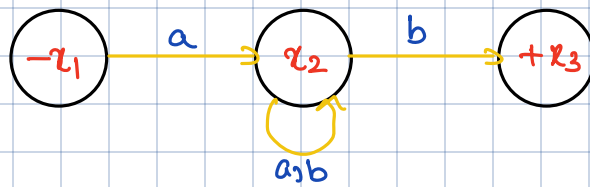
(iv)



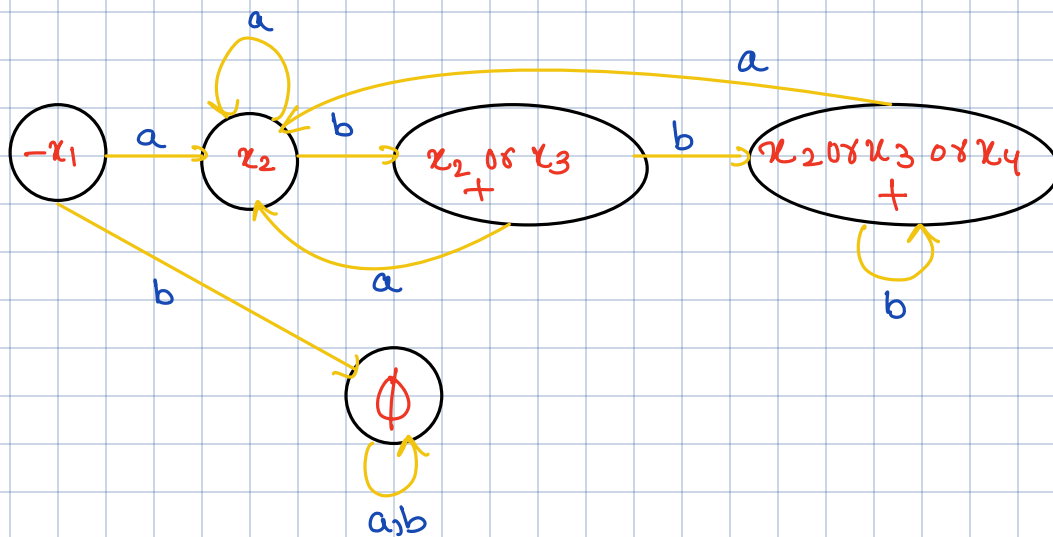
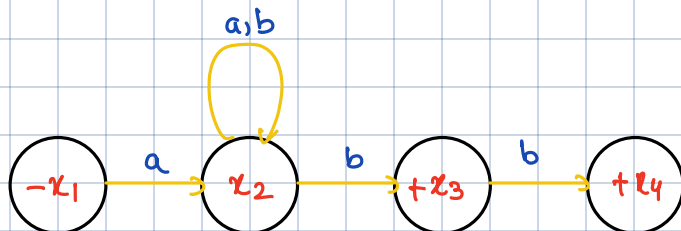
(v)



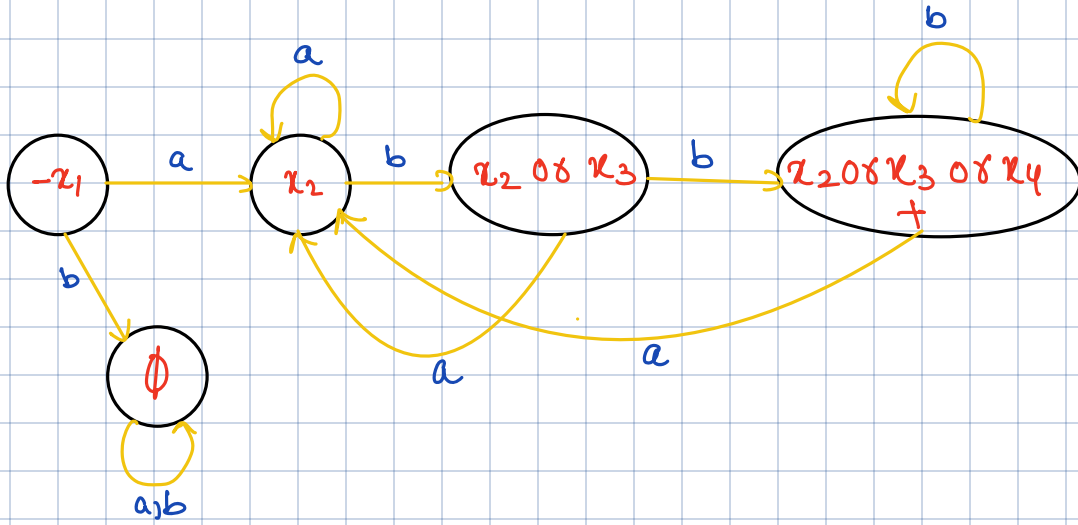
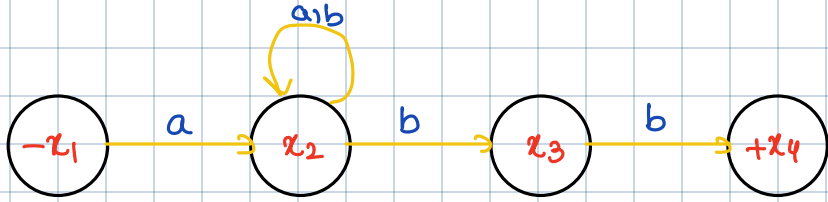
(vi)



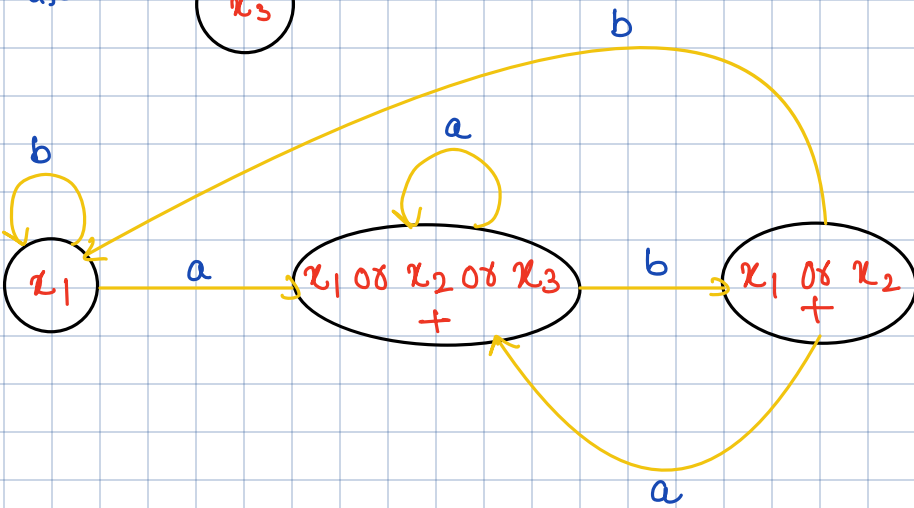
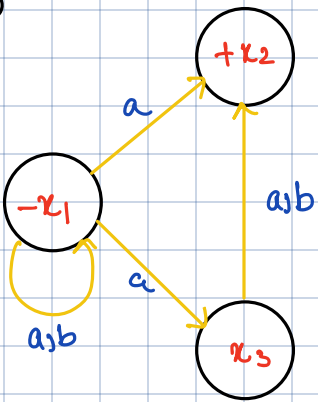
(viii)



(viii)

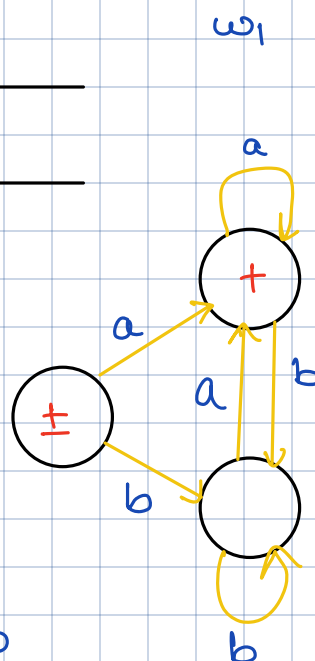
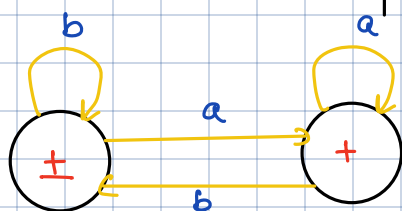


(ix)



Q6.

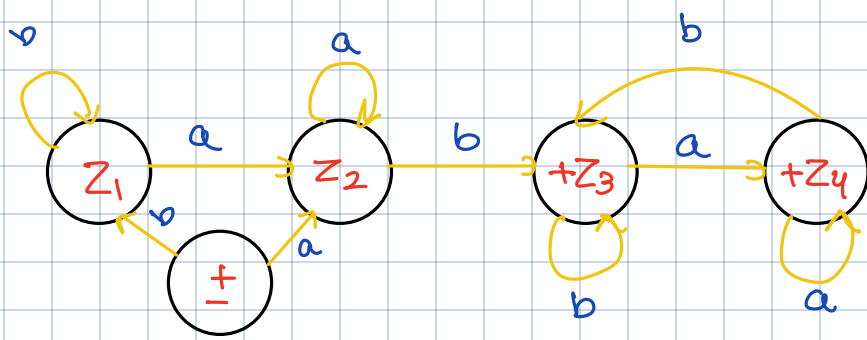
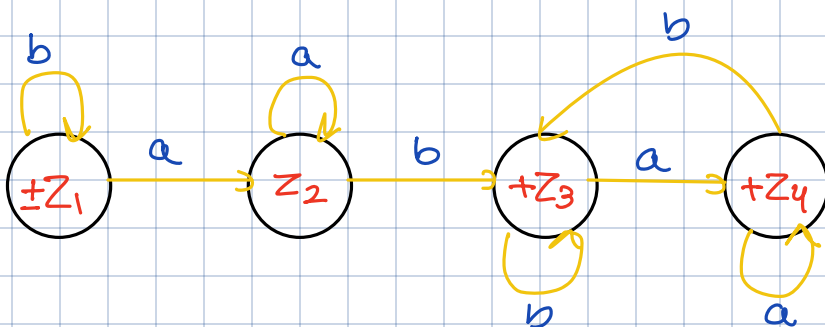
	a	b
$\pm Z_1 = \omega_1$	Z_2	Z_1
$+Z_2 = \omega_2 \oslash \omega_1$	Z_2	Z_1



(ii)

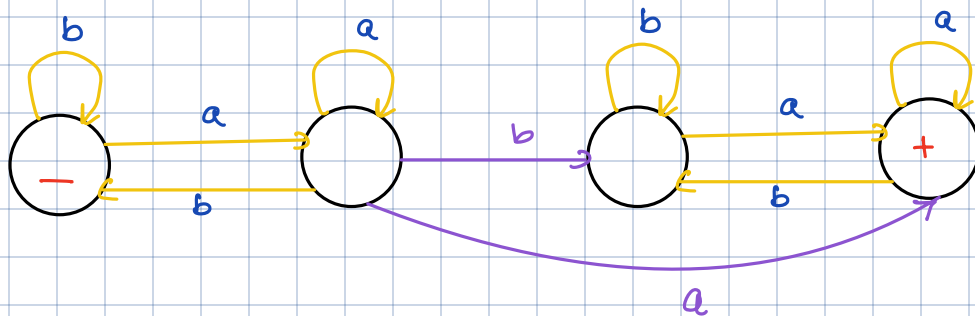
	a	b
$\pm Z_1 = \kappa_1$	Z_2	Z_1
$Z_2 = \kappa_2$	Z_2	Z_3
$+Z_3 = \kappa_3 \oslash \kappa_1$	$+Z_4$	$+Z_3$
$+Z_4 = \kappa_3 \oslash \kappa_1 \oslash \kappa_2$	$+Z_4$	$+Z_3$

$\kappa_3 \oslash \kappa_1$



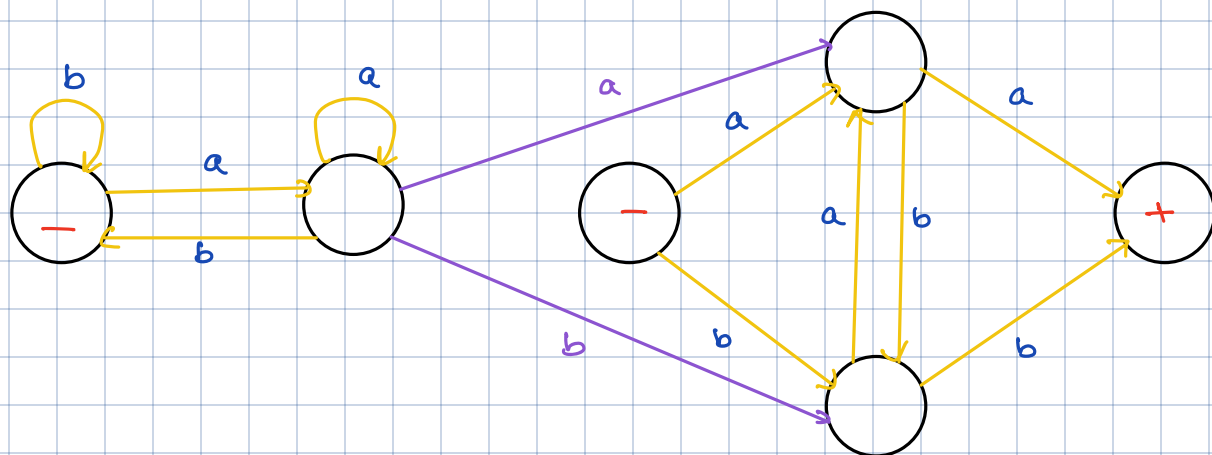
Q3.

(i)

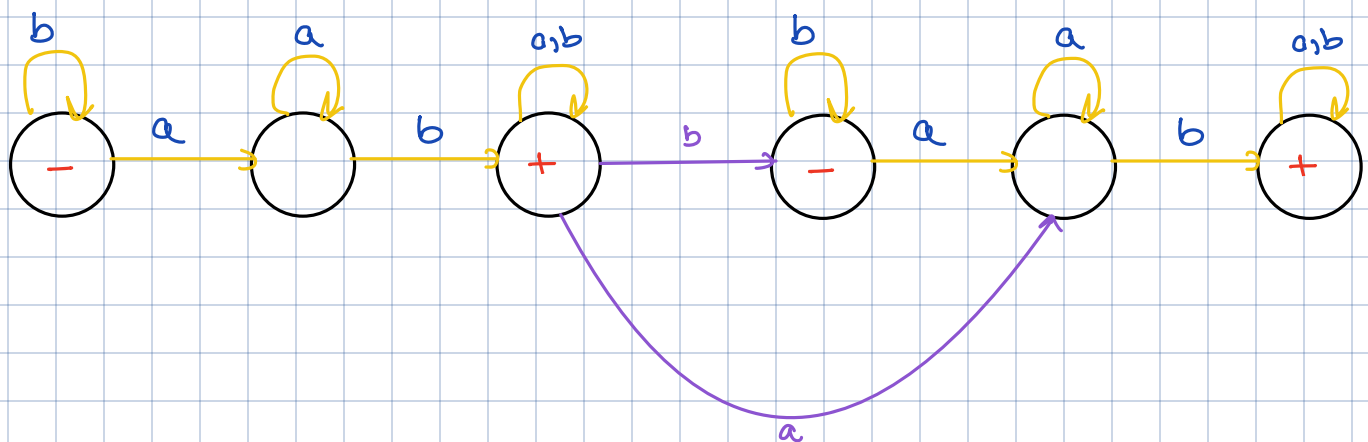


Construct FA₁, FA₂ by adding edges of FA₂'s - state on FA₁'s + state.

(ii)

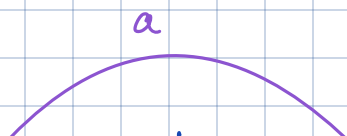


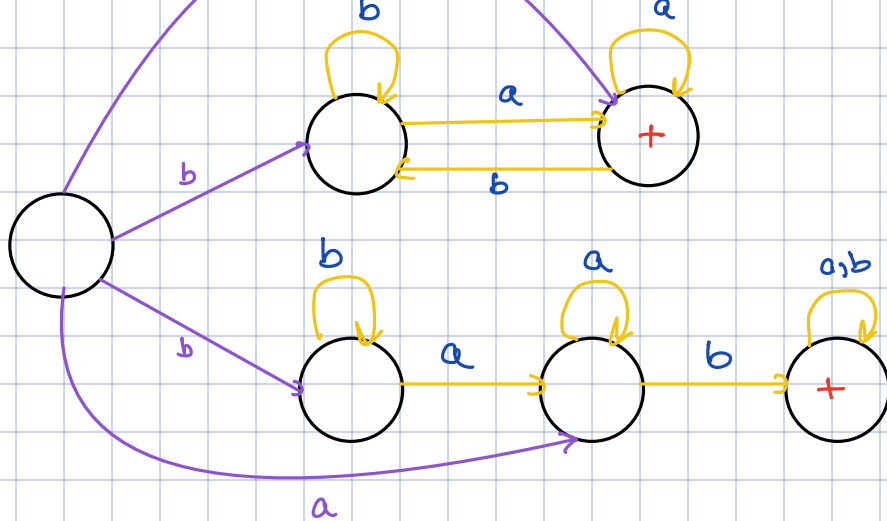
(iii)



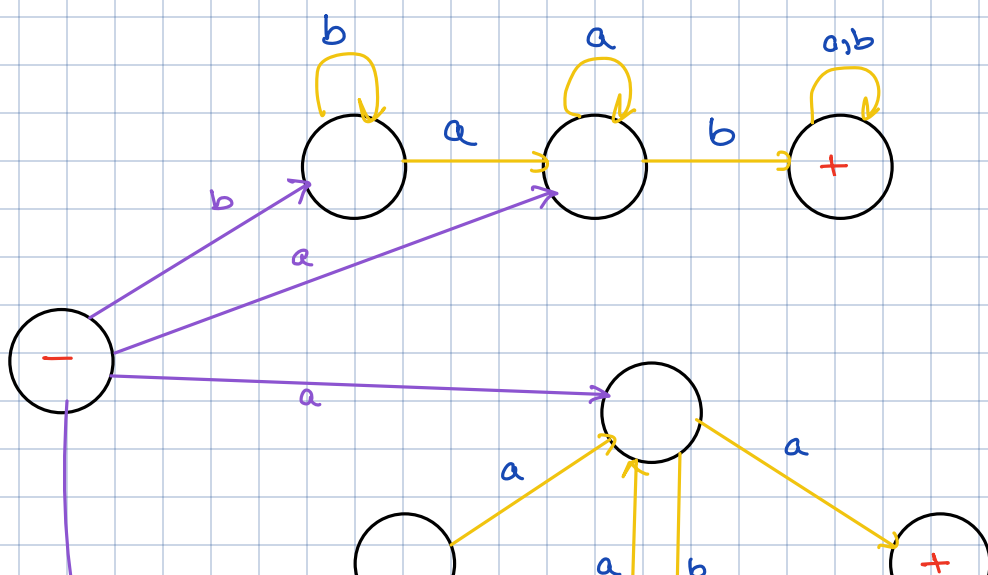
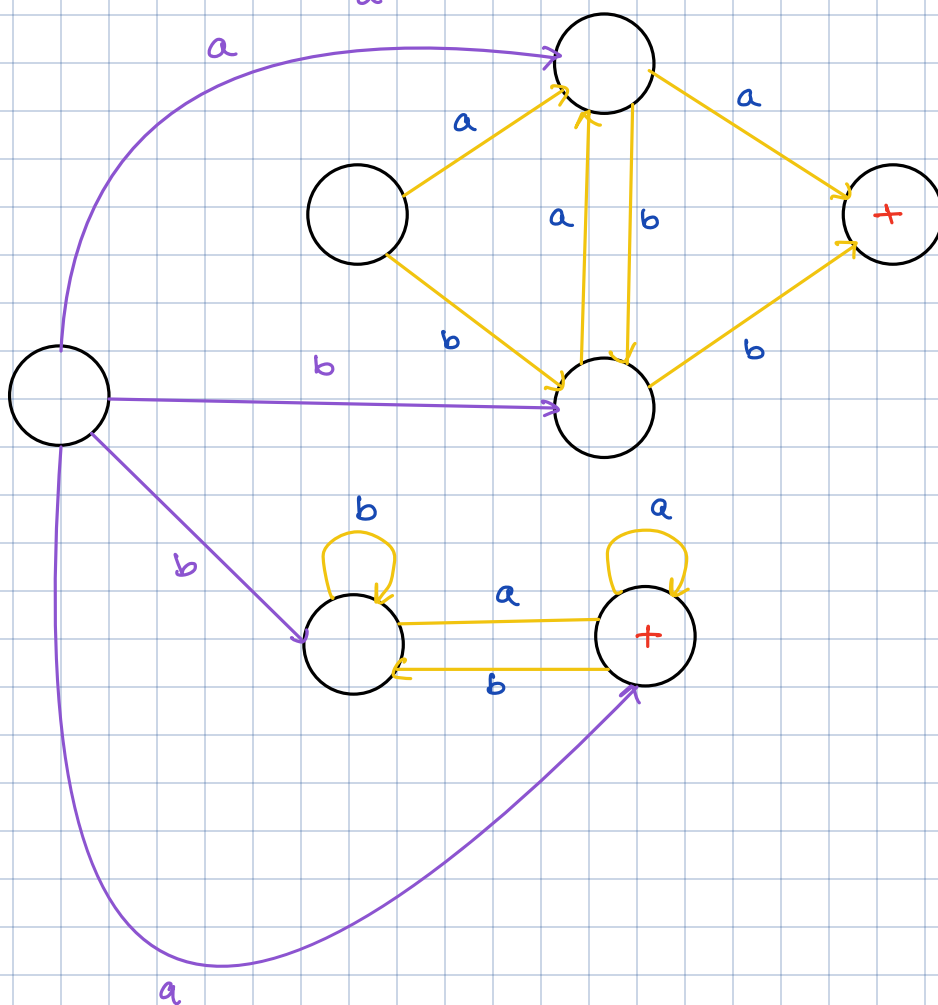
Q4.

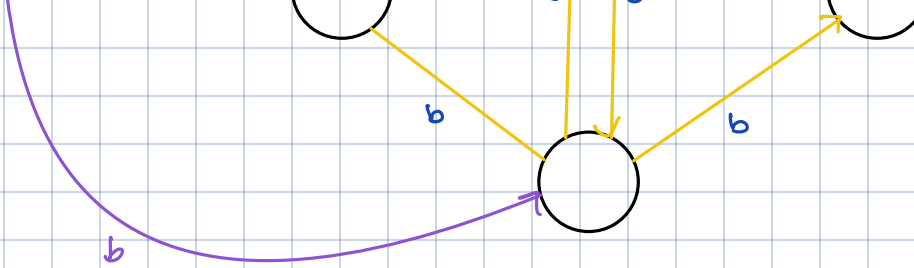
(i)



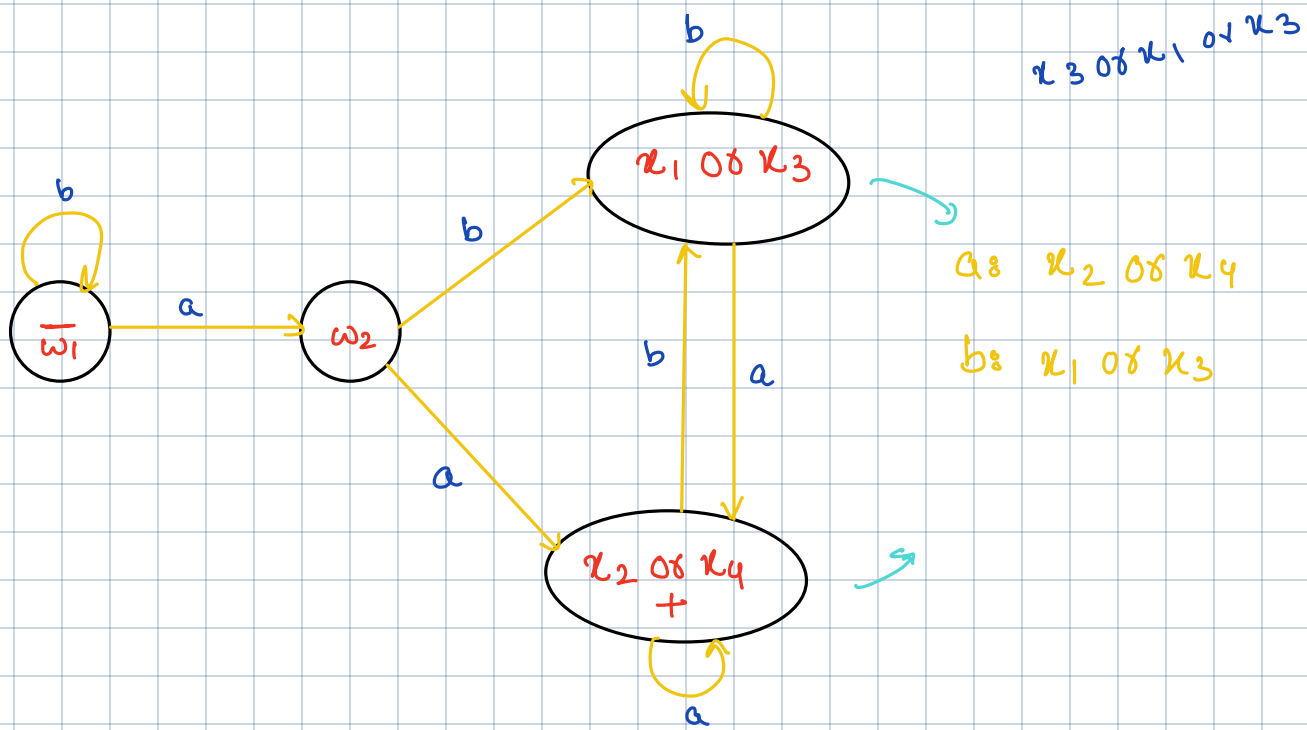
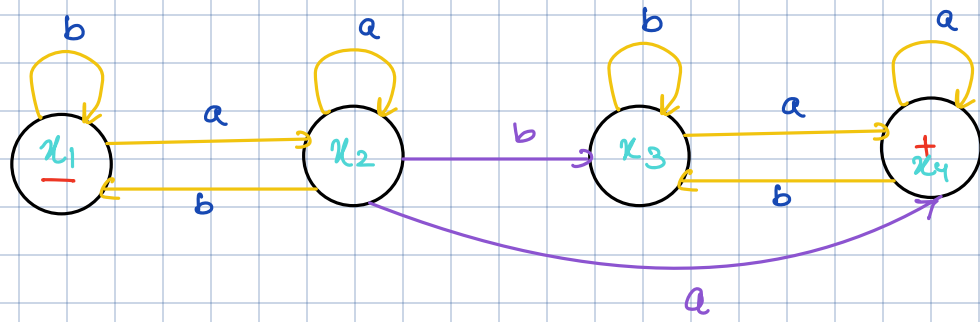


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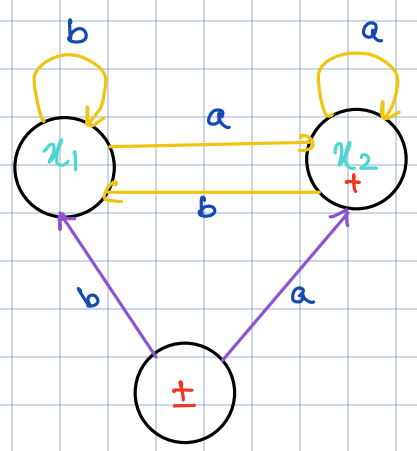




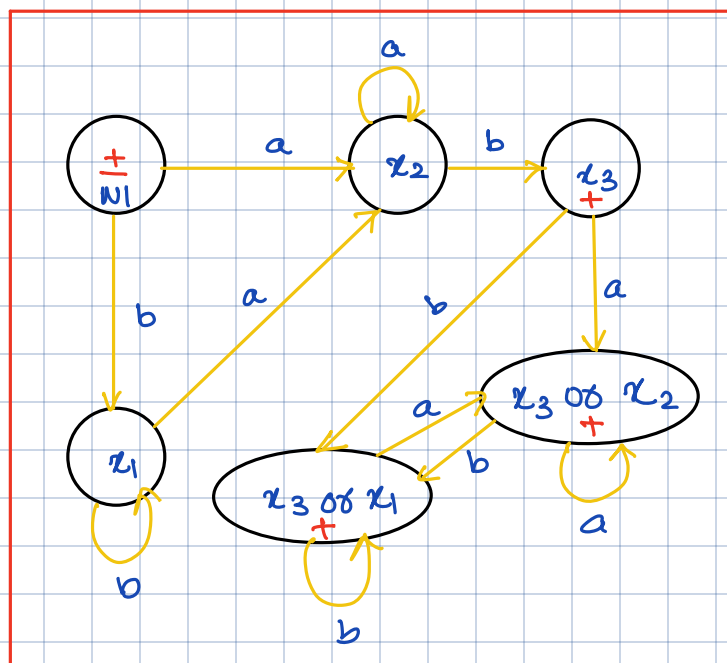
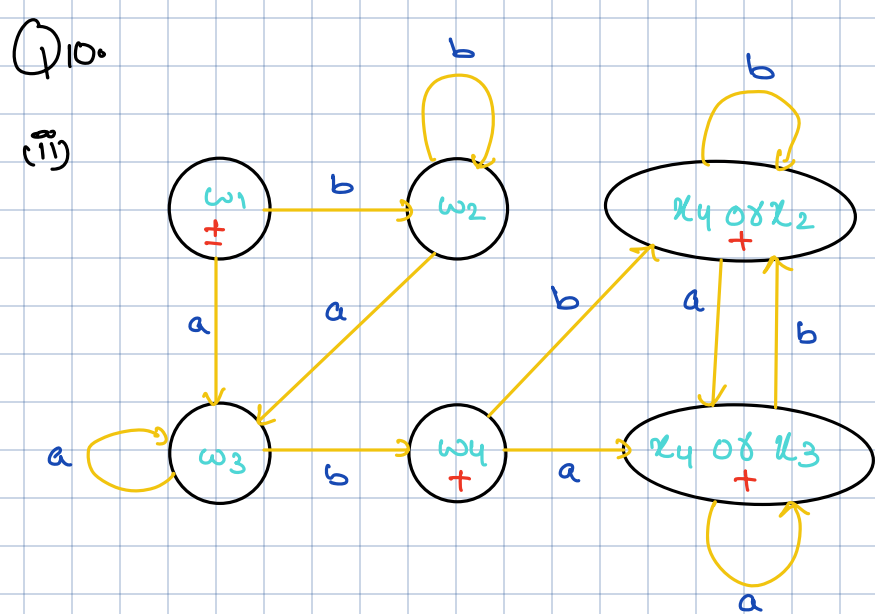
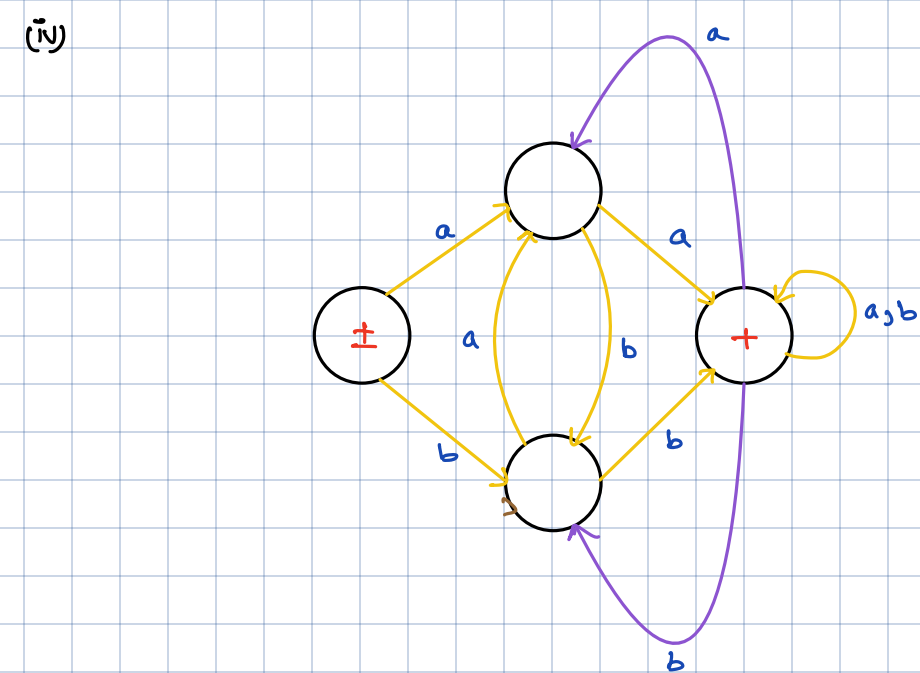
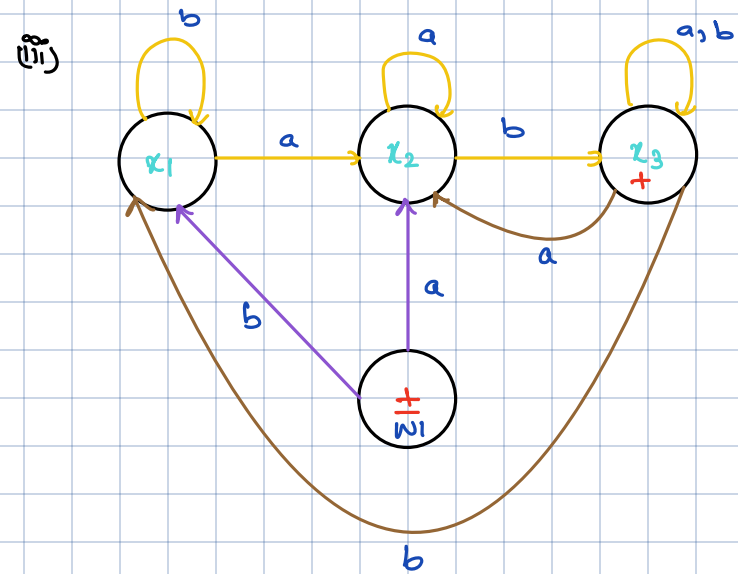
Q8.
(i)



Q9.
(ii)

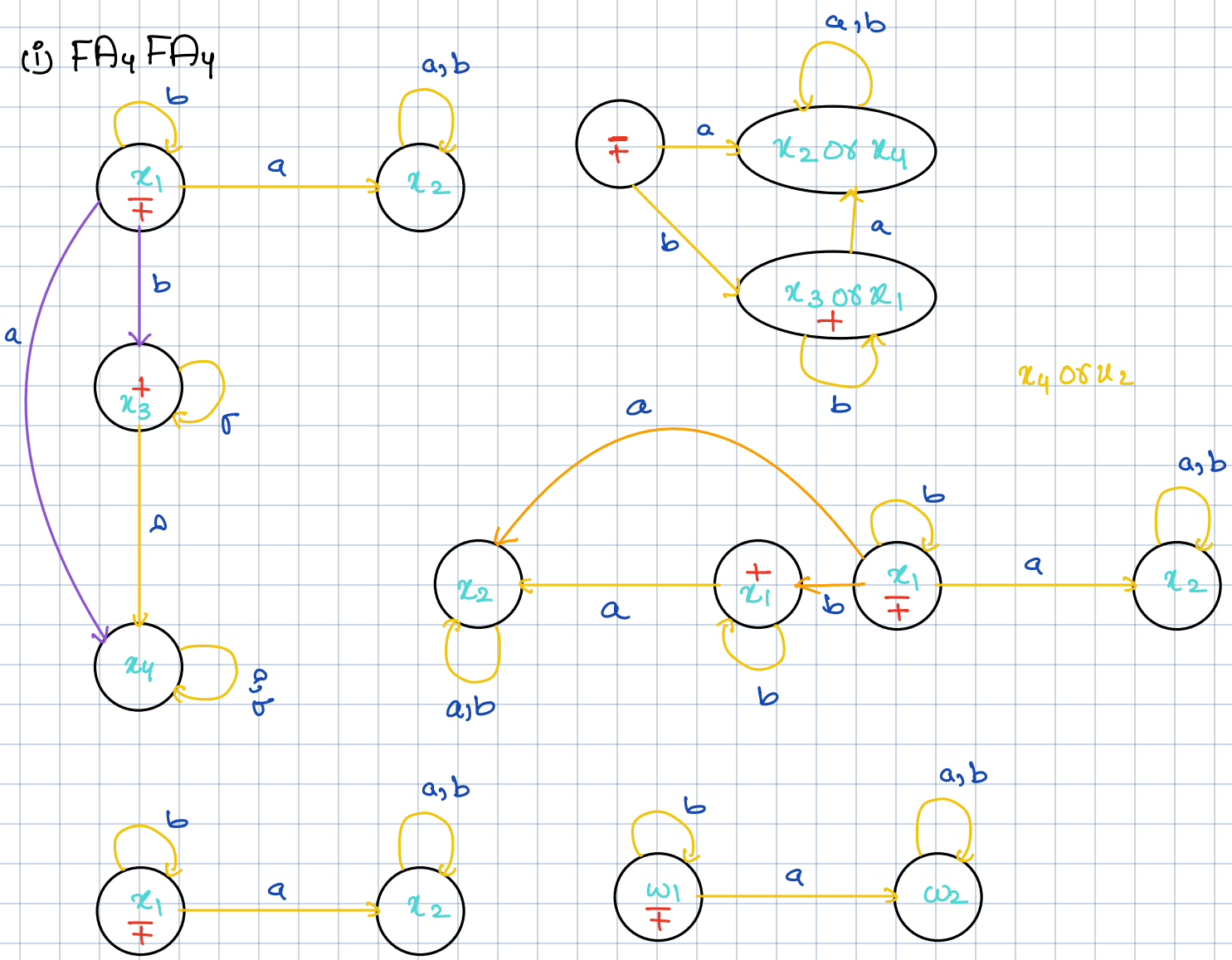


From every +, draw transitions to states by reading from '-'.
From every -, draw transitions to states by reading from '+'.

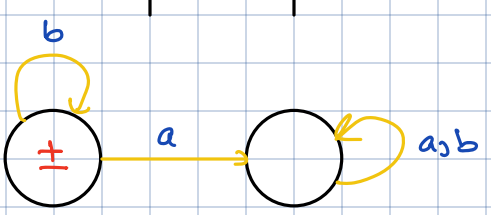


Q11.

(i) $FA_4 FA_4$



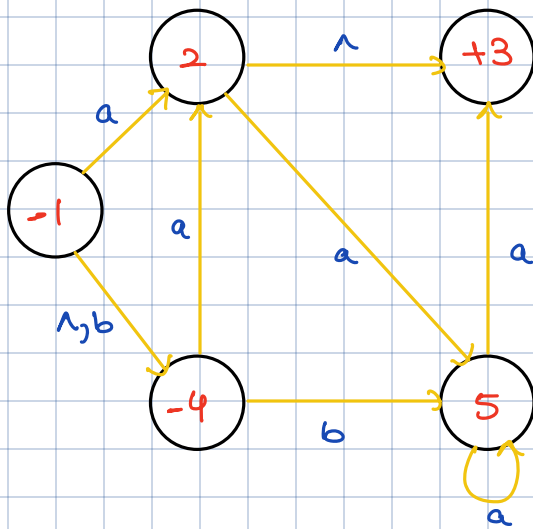
	a	b
$\pm Z_1 = x_1 \text{ or } w_1$		



check down

Q16.

i)



	a	b
A	B	C
+B	D	G
C	E	D
D	F	G
+E	F	G
+F	F	G

$$\epsilon\text{-closure}(\{1\}) = \{1, 4\} = A$$

$$\text{move}(A, a) = \{2\}$$

$$\epsilon\text{-closure}(\{2\}) = \{2, 3\} = B$$

$$\text{move}(A, b) = \{4, 5\}$$

$$\epsilon\text{-closure}(\{4, 5\}) = \{4, 5\} = C$$

$$\text{move}(B, a) = \{5\}$$

$$\epsilon\text{-closure}(\{5\}) = \{5\} = D$$

$$\text{move}(E, a) = \{3, 5\} \quad \text{move}(B, b) = G$$

$$\epsilon\text{-closure}(\{3, 5\}) = F$$

$$\text{move}(E, b) = G$$

$$\text{move}(C, a) = \{2, 5, 3\}$$

$$\epsilon\text{-closure}(\{2, 3, 5\}) = \{2, 3, 5\} = E$$

$$\text{move}(C, b) = \{5\}$$

$$\epsilon\text{-closure}(\{5\}) = D$$

$$\text{move}(D, a) = \{3, 5\}$$

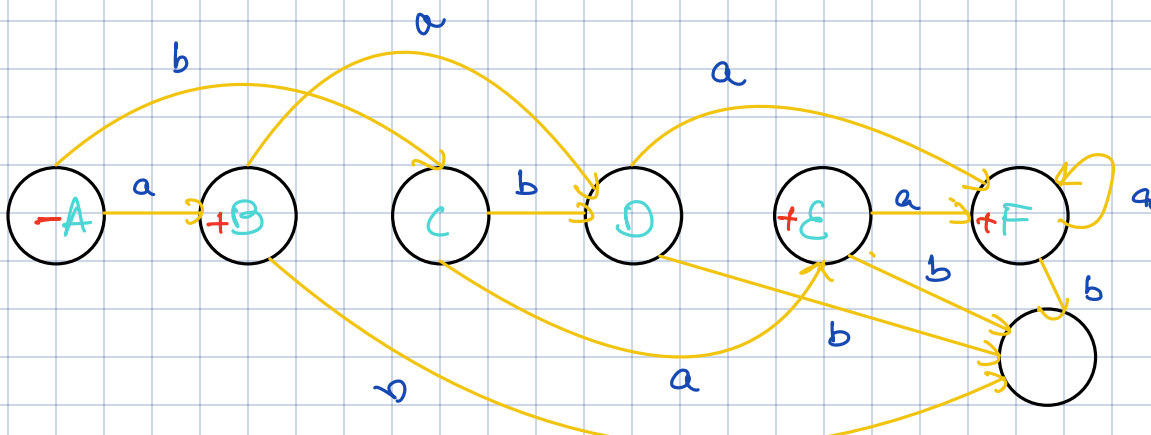
$$\epsilon\text{-closure}(\{3, 5\}) = \{3, 5\} = F$$

$$\text{move}(D, b) = G$$

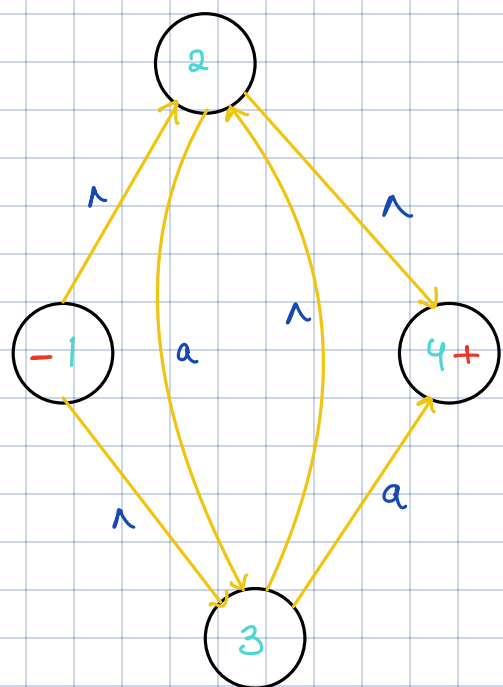
$$\text{move}(F, a) = \{3, 5\}$$

$$\epsilon\text{-closure}(\{3, 5\}) = F$$

$$\text{move}(F, b) = G$$



(ii)



	a	b
+A	B	ϵ
+B	B	ϵ

$$\epsilon\text{-closure}(\{1\}) = \{1, 2, 3, 4\} = A +$$

$$\text{move}(A, a) = \{3, 4\}$$

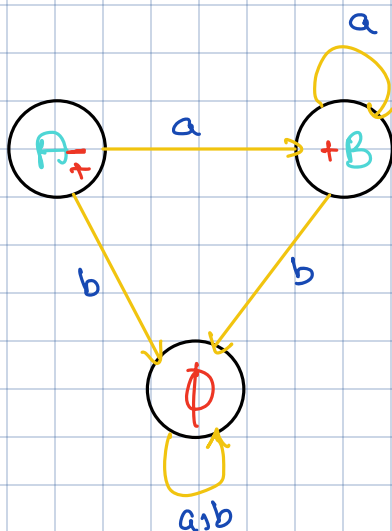
$$\epsilon\text{-closure}(\{3, 4\}) = \{3, 2, 4\} = B$$

$$\text{move}(A, b) = \epsilon$$

$$\text{move}(B, a) = \{3, 4\}$$

$$\epsilon\text{-closure}(\{3, 4\}) = \{3, 2, 4\} = B$$

$$\text{move}(B, b) = \epsilon$$



$$\epsilon\text{-closure}(\{1\}) = \{1, 3\} = A$$

$$\text{move}(A, a) = \{2, 3, 4\}$$

$$\epsilon\text{-closure}(\{2, 3, 4\}) = \{2, 3, 4\} = B$$

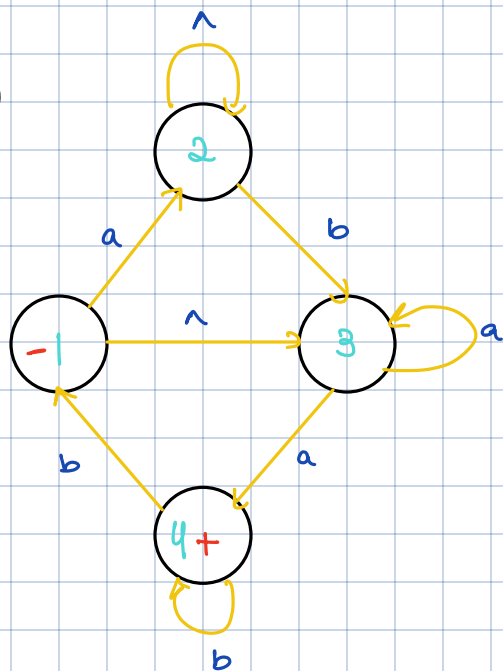
$$\text{move}(A, b) = \epsilon$$

$$\text{move}(B, a) = \{3, 4\}$$

$$\epsilon\text{-closure}(\{3, 4\}) = \{3, 4\} = C$$

$$\text{move}(B, b) = \{3, 4, 1\}$$

(iii)



	a	b
A	B	E
+B	C	D
+C	C	D
+D	B	D

$$\epsilon\text{-closure}(\{1,3,4\}) = \{1,3,4\} = D$$

$$\text{move}(D,a) = \{2,3,4\}$$

$$\epsilon\text{-closure}(\{2,3,4\}) = B$$

$$\text{move}(D,b) = \{1,4\}$$

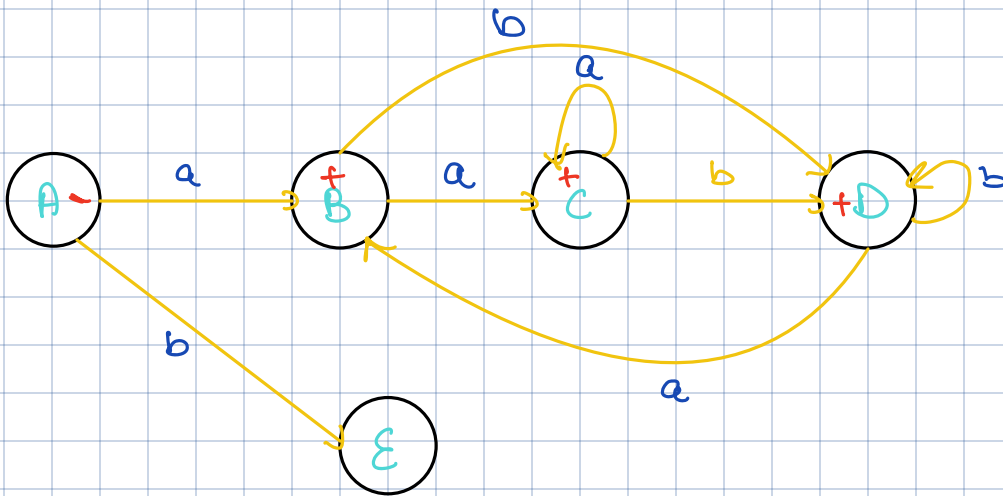
$$\epsilon\text{-closure}(\{1,4\}) = \{1,3,4\} = D$$

$$\text{move}(C,a) = \{3,4\}$$

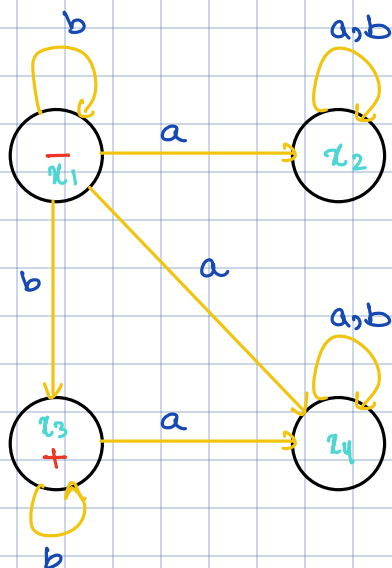
$$\epsilon\text{-closure}(\{3,4\}) = \{3,4\} = C$$

$$\text{move}(C,b) = \{1,4\}$$

$$\epsilon\text{-closure}(\{1,4\}) = \{1,3,4\} = D$$

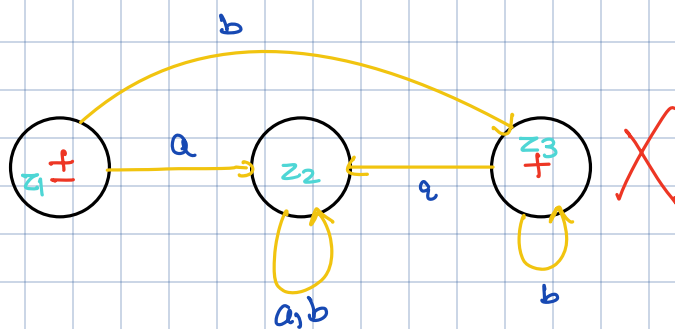


Q11.
(i)



	a	b
$\pm Z_1 = x_1$	Z_2	Z_3
$Z_2 = x_2 \text{ or } x_4$	Z_2	Z_2
$+ Z_3 = x_1 \text{ or } x_3$	Z_2	Z_3

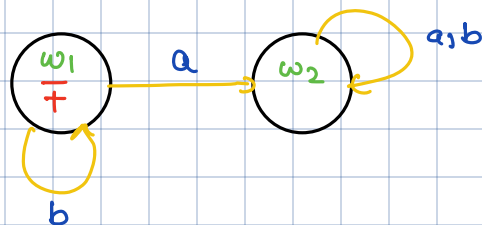
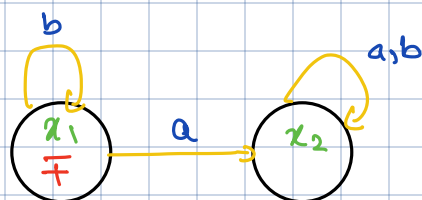
~~Below one~~



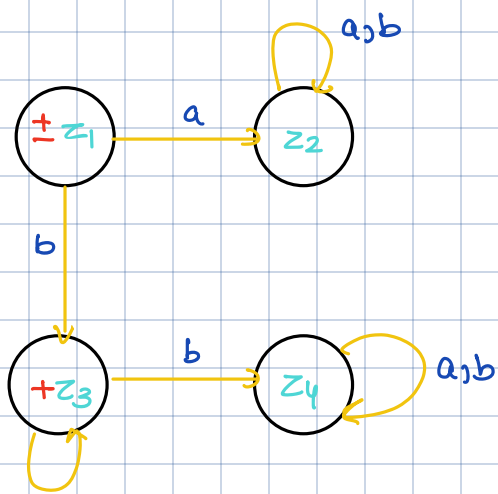
Q11.
(i)

	a	b
$-Z_1 = x_1$	Z_2	Z_3
$Z_2 = x_2$	Z_2	Z_2
$+ Z_3 = x_1 \text{ or } x_3$	Z_4	Z_3
$Z_4 = x_2 \text{ or } w_2$	Z_4	Z_4

Treat x_1 as $-$ & when you write letters to reach $(+)$ consider it $+$ & add $-$ of other FA here

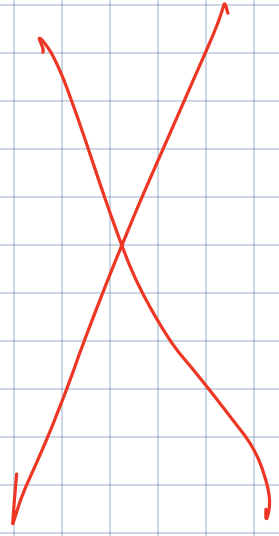
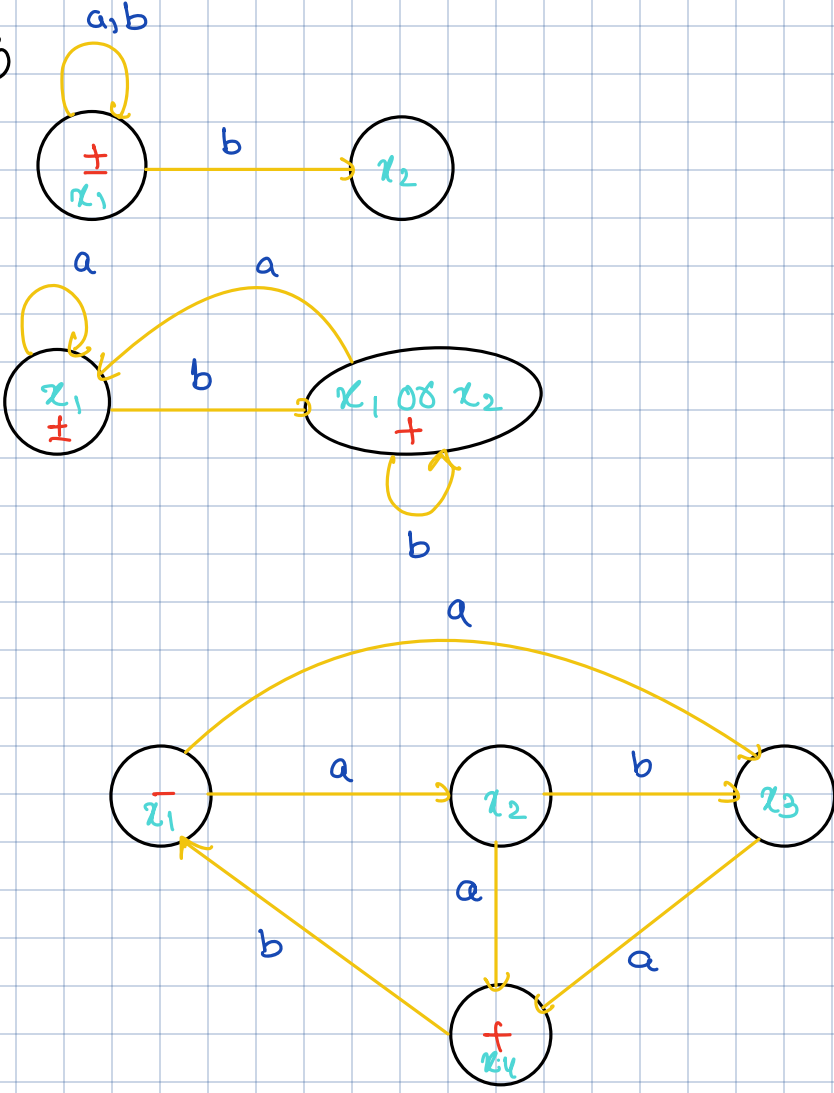


$Z_3:$
 $b:$
 $x_1 \text{ or } w_1 \text{ or } w_1$



b

(i)

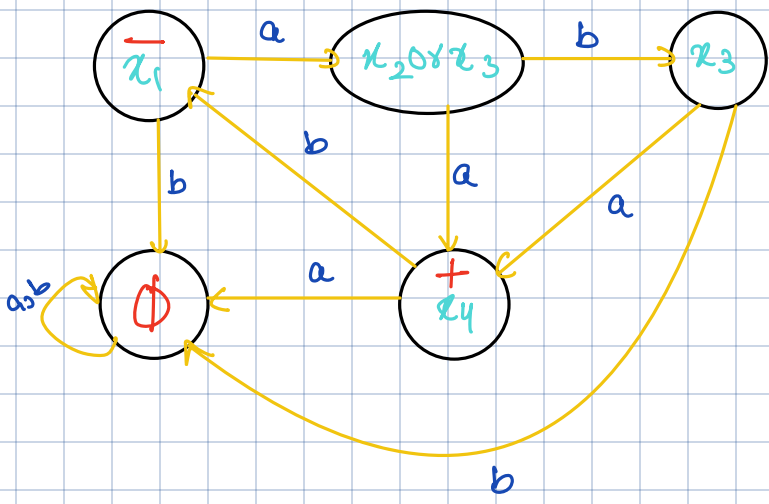


What is this?

aba

aa

aa



$$ab(a+b)^*ba + \Lambda + a + b$$

Handwritten annotations below the expression show red brackets and arrows indicating the decomposition of the expression into its constituent parts for the NFA construction:

- A red bracket under $ab(a+b)^*ba$ with an arrow pointing to the start of the main path.
- A red bracket under Λ with an arrow pointing to the transition from state 20 to state 13.
- A red bracket under a with an arrow pointing to the transition from state 1 to state 2.
- A red bracket under b with an arrow pointing to the transition from state 3 to state 4.

